

PLASTIC AND RECONSTRUCTIVE SURGERY

CHAPTER

1

Reconstructive Surgery

INTRODUCTION

Reconstructive Surgery is a branch of surgery that deals with the correction, restoration and improvement in shape and appearance of body structures that are defective, damaged, misshapen by injury diseases or growth. The goal of reconstructive surgery is “the restoration of form and function.” It is considered the core of plastic surgery.

The Reconstructive Ladder

It is a conceptual framework for understanding reconstructive options. It starts with the most simple option (healing by 2^{ry} intention) and progresses to more complex options in a step-wise fashion. When a wound or a defect cannot be closed primarily, and 2^{ry} healing is unlikely, too slow, or sub-optimal, grafts and flaps are used to restore normal function and/or anatomy.

INDICATIONS OF RECONSTRUCTIVE SURGERY

Congenital Lesions

- Cleft lip and palate.
- Vascular anomalies.
- Auricular deformities (e.g. prominent auricles).
- Hypospadias.
- Craniofacial conditions.
- Congenital hand anomalies.

Acquired Lesions

- Malignant lesions.
- Trauma
- Infection.
- Burns
- Miscellaneous.

GRAFTS

- A skin graft involves taking a healthy patch of skin from one area of the body, known as the *donor* site, and using it to cover another area where skin is missing or damaged. The piece of skin that is moved is entirely disconnected, and requires blood vessels to grow into it when placed in the *recipient* site.
- A graft *must* have a wound bed of healthy tissue (granulation tissue, muscle, fascia, bone with intact periosteum, or tendon with intact paratenon) because the graft is totally dependent on blood supply from the recipient site as shown in Figure 1.

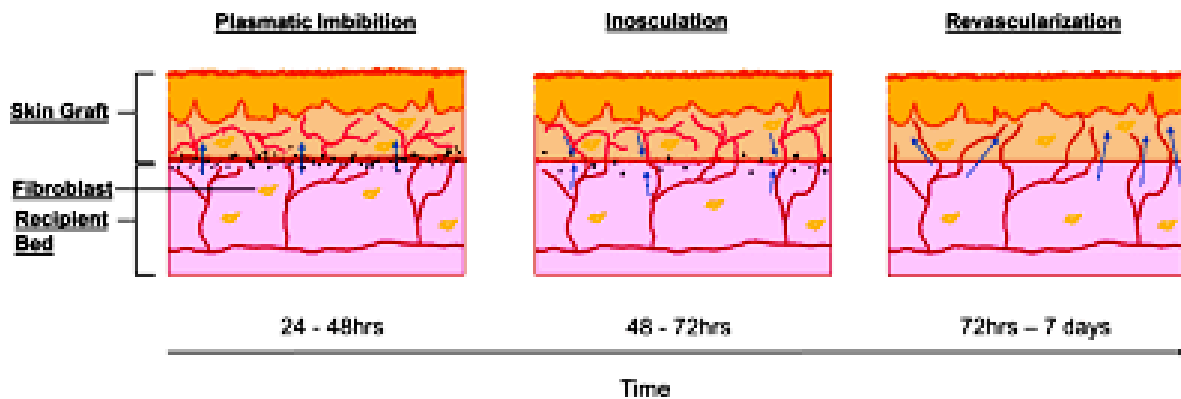


Figure 1: Vascularization of skin graft

Requirements of Graft Survival “Take”

- Well-vascularized bed.
- Immobile contact between the graft and the recipient.
- Low bacterial count at the site.
- Good general condition of the patient

Types of Skin Graft (Figure 2)

- Split-thickness skin graft (STSG) (*Thiersch* graft)
- Full-thickness skin graft (*Wolfe* graft)

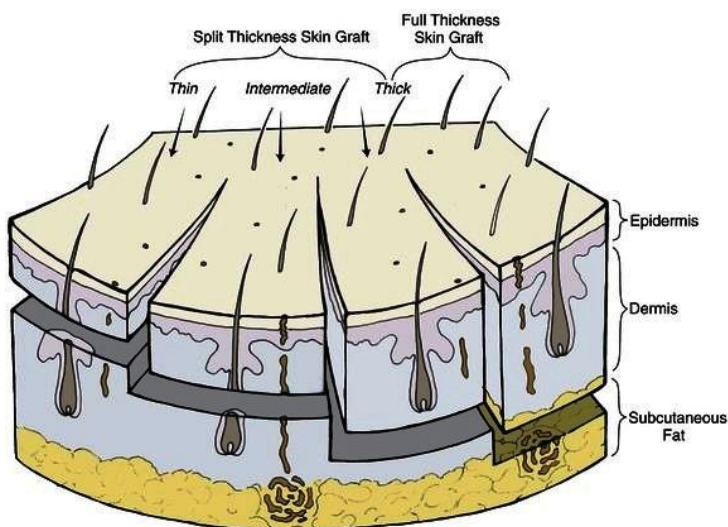


Figure 2: Skin and composite grafts. Note the split thickness skin graft and the full-thickness skin graft

The differences between Thiersch graft and Wolfe graft are summarized in Table 1

Table 1. Differences between Thiersch graft and Wolfe graft

Point of Difference	Thiersch graft	Wolfe graft
Thickness	Epidermis and varying degree of dermis. They are described as thin, intermediate or thick.	The entire epidermis and dermis.
Donor Sites	STSG leaves behind adnexal remnants such as hair follicles and sweat glands, foci from which epidermal cells can up-regulate and re-surface the donor site.	The donor site usually has skin redundancy. It is closed primarily.
Harvesting Technique	It is harvested with a skin grafting knife (Figure 3A), pneumatic or electric dermatome.	It is usually harvested with a scalpel (Figure 3B) in the plane between the dermis and subcutaneous (SC) fat.
Advantages	- Technically easy - Graft take-up is better - Donor site heals spontaneously	- Color match is good - No contracture - Sensation and function of sebaceous glands and hair follicles are retained better
Disadvantages	- Does not resemble normal skin (unpleasant scar) - Contracture - Least resistance to trauma	- Used for smaller areas - Limited donor sites - Requires more optimum conditions for survival
Indications	- Immediate coverage of clean soft tissue defects - Immediate coverage of burn defects and reduced fluid loss from the wounds - Prevention of contracture and enhancement of cosmesis in superficial wounds	- Small facial defects - Some types of finger defects - Reconstruction of donor sites of some flaps

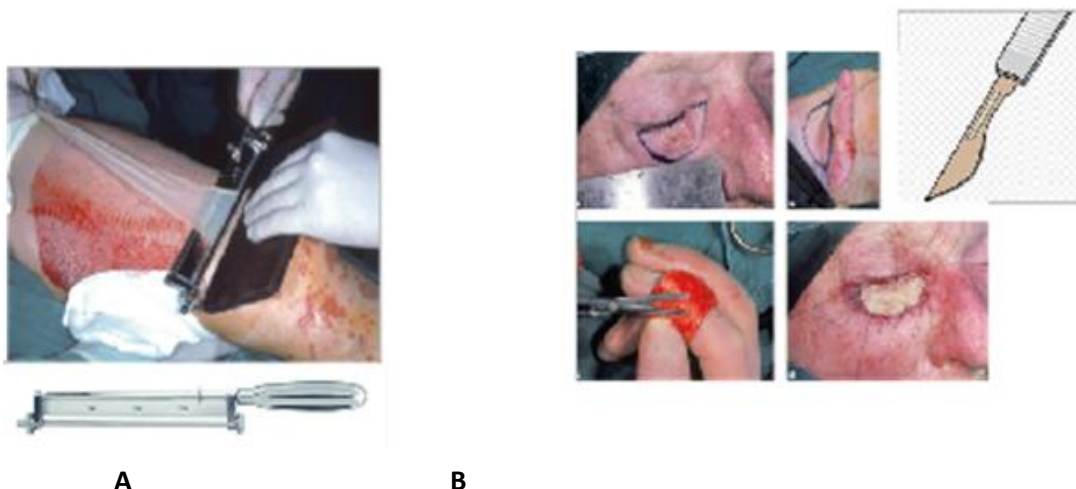


Figure 3: A: Skin grafting knife for Thiersch graft and **B:** Scalpel for Wolfe graft

Graft Nomenclature

- **Autograft:** It is tissue transfer from one location to another on the same patient.
- **Isograft:** Tissue transfer between two genetically identical individuals, eg, monozygotic twins.
- **Allograft (Homograft):** Tissue transfer between two genetically different members.
- **Xenograft (Heterograft):** Tissue transfer from a donor of one species to a recipient of another species.

FLAPS

Indications of flaps

- Flaps can be used when the wound bed is unable to support a skin graft (such as over exposed bone, cartilage, tendons, nerves, or hardware).
- To cover an area with poor vascularity.
- Full thickness loss of tissue: lid, nose, cheek, lip, and to cover bony prominences.
- For cosmetic reasons in the face.
- When later operations are needed in the recipient site.

Classifications of Flaps

- Flaps can be classified according to different parameters.
- A single flap can be classified according to more than one classification, and hence can have more than one nomenclature.

A. By Composition

Flaps can be classified by the type of tissue transferred into:

- *Single component*
 1. Skin flap- e.g. Parascapular flap
 2. Muscle flap- e.g. Rectus abdominis muscle flap or latissimus dorsi muscle flap
 3. Bone flap - e.g. Fibula flap
 4. Fascia flap -e.g. Temporo-parietal fascia
- *Multiple components* (Named by types of tissue)
 1. Fascio-cutaneous: e.g. Radial forearm flap or anterolateral thigh flap.
 2. Myocutaneous: e.g. Transverse rectus abdominis myocutaneous (TRAM) flap or latissimus myocutaneous flap.
 3. Osteo-cutaneous: e.g. Fibula flap with a skin paddle or medial femoral condyle flap with skin paddle.

B. By Location (Figure 4)

Flaps can be classified by the relation of the donor (**D**) and recipient (**R**) sites into:

1. Local flaps
2. Regional flaps
3. Distant flaps

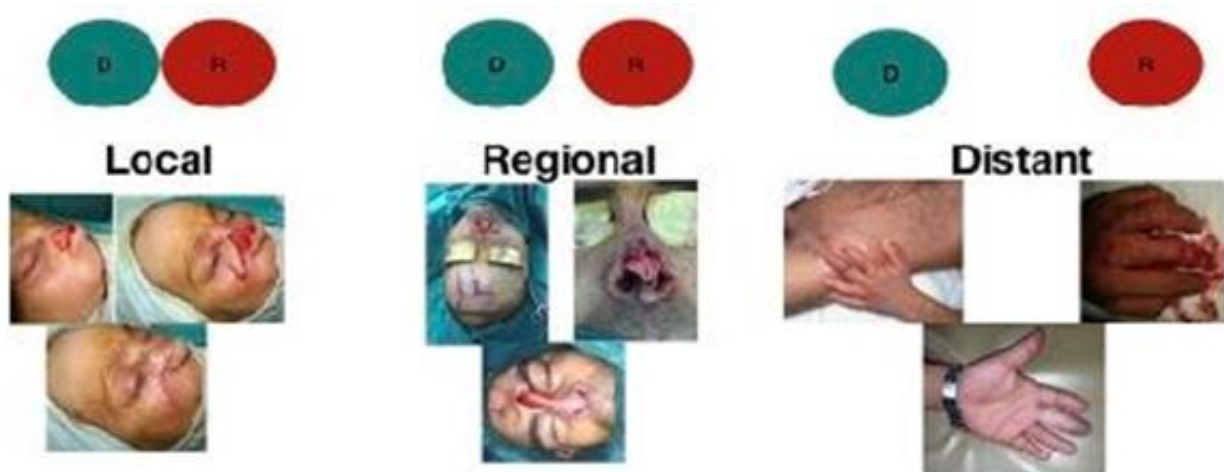


Figure 4: Classification of flaps by location into local, regional and distant flaps (according to relation of the donor (D) and recipient (R) sites)

C. By Vascular Pattern

Flaps can be classified according to the vascular pattern into

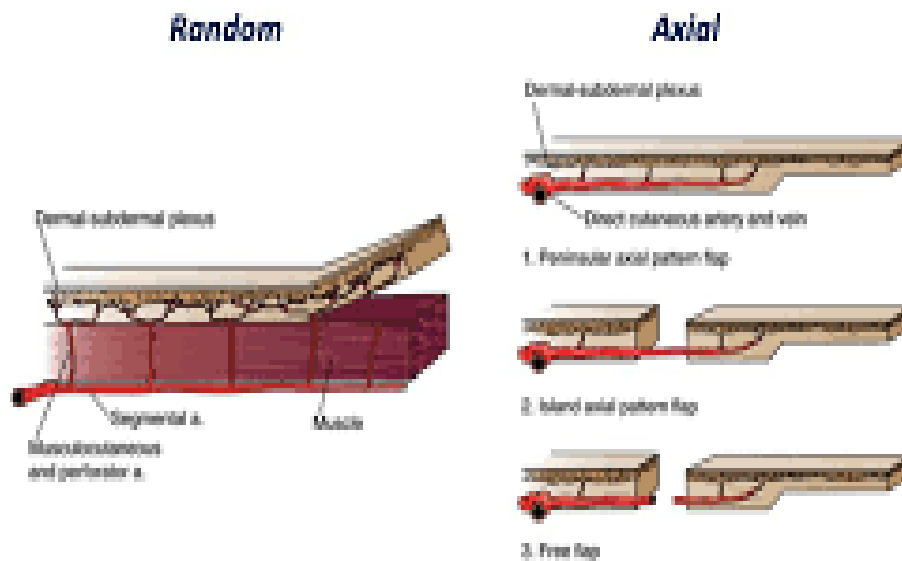


Figure 5: Random pattern and axial pattern flaps

1. Random pattern

- Do not have a specific or named blood vessel as their blood supply
- Instead, these flaps are designed based on the size of the flap base
- They are therefore limited in dimensions specifically in length : width-base ratio of 1:1 (may be larger in the face)
- If designed in a larger ratio, the random blood supply often cannot support the flap.
- Example: Rhomboid flap (Figure 6)



Figure 6: Rhomboid flap is a type of random flaps

2. *Axial Pattern*

- Designed with a specific named blood supply that enters the base and runs along its axis
- This allows the flap to be designed as long and as wide as the territory the axial artery supplies (angiosome)
- Blood supply requires an artery and its accompanying vein
- Greater length is possible than with a random flap
- All free flaps are axial
- Penninsular (skin and vessel intact in pedicle)
- Island (vessel intact but skin over the pedicle)

Burn Injuries and Management

INTRODUCTION

Burn injury of the skin is characterized by the damage to skin tissue from hot (scald, flash, flame, contact), cold, electrical, chemical, radiation, sunlight, or other sources. Burns constitute one of the most common causes of morbidity and mortality worldwide. They can result in significant disfigurement, physical impairment, work loss, psychological problems, and considerable economic burden. Prevention of burn is considered the best strategy to reduce the overall burden of burns. The impact and the management of burn injury depend on the severity of burn. Although minor burns can be treated at outpatient clinics, the management of patients with severe burns requires multidisciplinary approach in specialized burn care centers.

Burn trauma differs from the other causes of injuries in many aspects. Increased knowledge about the pathophysiology of burn provided better treatment plans and led to the improvement of overall outcome for these patients. Formation of scar is an undesired consequence of burn with many long-term complications. The local treatment of burn wound should address the major concerns of wound care including anti-inflammatory treatment, wound coverage, and prevention of infection and scar formation. Although superficial burns may be managed with topical treatment, deep burns require excision and grafting. As traditional treatments have many limitations, alternative options with better outcomes have been searched in the restoration of damaged tissues. Tissue-engineered products, stem cells, and gene therapy constitute new concepts that offer promise in the treatment of burn wounds. Although the results with these innovations are encouraging, they require sophisticated techniques, and evidence for their long-term efficacy in burn wounds is lacking. Future search will introduce novel therapeutic options and assist in the establishment of standard burn wound care in clinical settings

1. EPIDEMIOLOGY AND RISK FACTORS OF BURNS

Burns constitute a major health problem worldwide. Considerable amount of patients suffer or die from burn injuries globally. The burns mostly occur in low- and middle-income regions of the world. Burn injuries occur more commonly in men at young adult age; however, in the elderly, female predominance is seen. Alcohol usage, smoking, presence of open fire source or ground level stoves, wearing high-risk cloths (long, loose-fitting, synthetic), improper temperature setting of water heaters, use of unsafe electrical equipment or kerosene lamps, low socio-economic status, over-population, illiteracy, unemployment,

belonging to a large and single-parent family, and housing without adequate health and safety requirements are all reported to be risk factors for burn injury

SEVERITY OF BURN INJURIES

Fortunately, most of the burn injuries fall into mild cases that can be treated in community or in outpatient clinics. However, depending on the severity of the condition, hospitalization or treatment in intensive care unit (ICU) may be needed.

Severity of a burn injury depends on the extent of burned area (expressed as the percentage of total body surface area (TBSA)), depth of tissue damage, presence or absence of inhalation injury, mechanism of injury, age of the patient, and accompanying comorbidities. Median TBSA of all burn cases was reported as 15%, and severe burn injuries constitute <10% of total burns. Mostly children, women, and elderly people are affected by severe burns. Low socio-economic status and being from ethnic minorities are considered as risk factors for experiencing severe burns. Inhalation injury is seen in < 4% of cases and more likely to be observed in extensive burns.

ETIOLOGY OF BURN INJURIES

Burn injuries can result from diverse etiologies including flames, scalds, contact, electricity, chemicals, or even sunlight. The mechanism may differ according to the sex, age, residence, ethnicity, and admittance status (admitted or non-admitted) of the patient. In general, scald, flame, and contact are the major mechanisms for burns. Electrical and chemical burns occur less frequently. Other than the above-mentioned mechanisms, many other causes including sunburn and flash lasers can also result in burn injury.

MORTALITY FROM BURN INJURIES

Mortality rate from burn injuries differs among different studies and is reported between 1.4 and 18%. Older age, high extent of burned surface, concomitant illnesses, the presence of inhalation injury, African-American race, urban practice setting, and facial location of burn are all considered as risk factors for mortality. Flame burns are in general more fatal than contact burns. Mortality from burn injury is most commonly related to multi-organ failure and sepsis. Pneumonia and acute respiratory distress syndrome (ARDS) are also associated with mortality.

ETHICAL ISSUES

In all, but especially pediatric and elderly burns, legal and ethical issues should be considered. As abuse and mal-treatment may go unnoticed, identification of suspicious injuries by the physician is important. Delayed referral, suspicious and unreliable history, inconsistent explanations of parents or caregivers, tap water injury, and the presence of immersion lines are some of the clues that should raise the suspicion of abuse.

PRECAUTIONS FOR BURN PATIENTS

As most of the burns occur accidentally, prevention strategies remain the best approach in order to reduce the morbidity and mortality associated with burns. Increasing knowledge about the epidemiology of burn injuries will aid in defining preventable risk factors that should be targeted. While safety interventions for work-related burns decrease the risk, certain cultural practices and social habits may be related with increased burn accidents in certain geographic regions. Although education and increased awareness of public play important roles in prevention strategies, the introduction of legislation and better regulations are more effective in reducing the burn injury. Additionally, enforcement of legislation is critical to increase the success of prevention programs

PATHOPHYSIOLOGY OF BURN WOUNDS

There are various models for burn wound evaluation. One of most commonly used is *Jackson's model* in which 3 concentric areas can be detected based on the severity of tissue damage and changes in blood flow of a burn wound. Briefly, the first zone is the zone of coagulation; this is the point of maximum damage with irreversible tissue loss due to coagulation of the proteins and tissue necrosis. Surrounding the coagulation zone is the zone of stasis, which is characterized by decreased perfusion. This ischemic zone may progress to full necrosis unless the ischemia is reversed. Therefore, the main aim of burn resuscitation is to increase tissue perfusion here and prevent any further damage. The outermost layer is the zone of hyperemia. Tissue perfusion is increased and the tissue here invariably is recovered, unless there is severe sepsis or prolonged hypoperfusion

Systemic nature of the burn injury is unique that should be taken into consideration while approaching the patient. Understanding the pathophysiology of burn will provide useful information for early and effective management of burn patients, improve the quality of care for burn wounds, allow the identification of novel targets for the treatment of scar formation, and contribute to efforts to reduce the mortality. The local burn wound induces a generalized inflammatory response characterized by the activation of cytokines and release of various growth factors that can result in detrimental effects on many organs. The magnitude of this response depends on the severity of burn. One of the distinct features of burn injury is that the cytokine-mediated signaling triggered by the tissue damage results in a generalized increase in capillary permeability and extravasation of plasma causing exaggerated edema response even at distant sites (Figure 1).

Loss of intra-vascular fluid is accompanied by a decrease in cardiac output and increase in peripheral vascular resistance that may lead to hypoperfusion of organs and burn shock. Hypercoagulability may occur due to systemic activation of platelet aggregation and fibrinolysis. After the edema phase, a hyper-metabolic state ensues which is characterized by an increase in oxygen consumption, marked protein and lipid catabolism, increase in energy requirements, high cardiac output, tachycardia, severe muscle weakness, cachexia, and decrease in immune functions

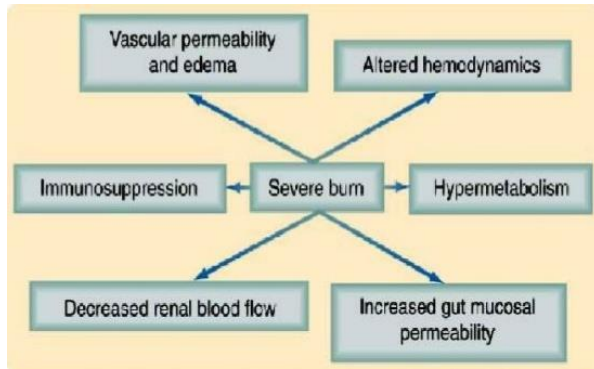


Figure 1: Summary of the pathophysiology of burn

Although the wound healing phases are similar to other types of wounds, the prolonged healing time is especially important in burn wounds. The severity of burn, the mechanism of injury, and associated diseases of patients influence wound healing. The *inflammatory phase* includes the vasodilatation and inflammatory cell migration through the cytokine signaling cascade. In the *proliferative phase*, epithelization takes place by the migration of keratinocytes from the epithelium of the wound edges and dermal appendages. The *remodeling (maturation) phase* is characterized by the deposition of collagen by myofibroblasts, compaction of the connective tissue, and finally the contraction of the wound. Although the wound contraction and scar formation are normal and necessary for the closure of wound, excessive fibrosis and increased tensile stress during remodeling carry the risk of abnormal scar formation. Intense and prolonged inflammatory response with increased release of cytokines, growth factors, and other mediators from the inflammatory cells and platelets are associated with scar formation. The depth of burn, age of the patient, the treatment, and response of wound are important determinants for the development of scar tissue. Wounds that are not healed in 2–3 weeks are generally at risk of developing aberrant scar tissue.

Superficial burn wounds heal completely in 5–7 days during the proliferative phase. As the required dermal components are lost in deep burns, proliferation cannot be provided, and the epithelialization is delayed. The lack of supportive and vascular tissue is associated with abnormal contraction, and these wounds heal with hypertrophic scarring and contractures if left to heal spontaneously.

MANAGEMENT OF THE BURN PATIENT

Early and appropriate treatment of burn injury is associated with better prognosis. Pre-hospital management and the treatment of burn patients in the emergency department fall out of the scope of this chapter and follow the general rules for trauma patients. As the airway edema may start soon after burn and unexpectedly, early intubation may be indicated. Since massive edema may develop in extended burns, all jewelry and accessories should be taken off. Specific interventions may be indicated according to the mechanism of burn (electrical, chemical burns). Until the patient is referred to the medical center, wounds should just be covered with clean cloth. Cooling with compress may be done; however, unburned regions should be kept warm in order to avoid hypothermia.

As the *hypovolemic shock* is associated with high morbidity and mortality, *fluid resuscitation* should be done early and adequately. Several criteria have been described for fluid resuscitation of burn patients

Evaluation of the Severity of Burn and Referral of the Patient

Extent of the Burn

Assessing the severity of burn injury is important in deciding the need for hospitalization. To assess the *extent of burned area*, several methods can be used including rule of nines, rule of fives, and Lund-Browder Chart, which is especially used for children as it more accurately estimates the age-specific percentage of TBSA. Additionally, for local assessments of small burns, the palm method (palm with fingers accounts for 1% of TBSA) can be used practically in adults. First-degree burns are not considered in the calculation of TBSA.

Depth of the Burn

Although more precise methods including laser Doppler flowmetry and video microscopy have been defined, the depth of burn is mostly estimated clinically (the presence of pain or blister, appearance and color of the skin) in practice. Burns can be classified into three types according to the depth of injury:

1. *First-degree (superficial)*: Only the epidermis is involved. The skin is red and painful. There is usually no blistering and skin will blanch when touched. It heals without scarring.
2. *Second-degree (partial thickness)*: In superficial dermal burns (SDB), only the papillary dermis is involved. In this case the skin is painful. Blisters are seen. When the bullae are de-roofed, the skin is wet and blanches when touched. It heals with minimal pigmentary changes without hypertrophic scarring. If the reticular dermis is involved, it is considered as deep dermal burn (DDB). In this case, there is less pain and blistering. There is eschar and the skin is mottled white. It heals with scarring.
3. *Third-degree (full-thickness)*: There is little or no pain. It involves the epidermis and dermis and extends to the subcutaneous layer. The skin is leathery white or dark, and inelastic. There is eschar. It does not blanch. It does not heal spontaneously, results in hypertrophic scar and contractures, and requires grafting.

Patient Referral Criteria

After the pre-hospital stabilization of the patient, depending on the severity of injury, treatment in a more equipped hospital may be required. The referral criteria may vary across different studies and include:

1. Partial-thickness burns >10% of TBSA in patients who are <10 years or >50 years of age
2. Partial-thickness burns >20% of TBSA in other age groups
3. Burns that involve the face, hands, feet, genitalia, perineum, or major joints
4. Third-degree burns in any age group
5. Electrical burns, including lightening injury
6. Chemical burns

7. Inhalation injury
8. Burns in patients with pre-existing medical disorders that could complicate management, prolong recovery, or affect mortality rate
9. Any patients with burns and concomitant trauma (such as fractures) in which the burn injury poses the greatest risk of morbidity or death
10. Burn injury in patients who will require special social, emotional or long-term rehabilitative intervention

Management Policy

While major burns must be managed in hospitals with multidisciplinary burn teams, moderate burns can be managed in minor hospitals. On the other hand, minor burns can be treated at outpatient clinics.

In burn patients who require intravenous (IV) *resuscitation*, a Foley catheter is placed early so that urine output (UOP) can be monitored as a guide for volume status. At this time, a nasogastric (NG) tube may also be inserted to decompress the stomach and begin early enteral feedings as part of the resuscitation recommended by the American Burn Society. Peripheral pulses are assessed immediately, and all extremities and the chest wall are evaluated for potential compartment syndromes. Initially, weak pulses are assumed to result from under-resuscitation, but a low threshold to perform escharotomies or fasciotomies should be maintained, especially in patients who are transferred from outside facilities some hours after the event occurred

Pain management is important in burn patients since the discomfort from pain results in anxiety, increases the risk of prolonged hospitalization, leads to loss of patient confidence, and complicates the interventional procedures. Burn patients may suffer from different types of pain including background and procedural pain. In severe burns, moderate to potent opioids (fentanyl, morphine, ketamine, and others) are preferred, and non-steroidal anti-inflammatory drugs (NSAIDs) may be added to reduce the overall dose of opioids. NSAIDs may be sufficient to relieve pain in patients with mild to moderate burns. Anti-depressants and anti-convulsants are used as first-line therapies for neuropathic pain that may be seen in burn patients. Psychological therapy has also been reported with various successes for management of pain.

Resuscitative Fluid Management

Formulas and Solutions

Historically, fluid management for burns has been as much an art as it has been a science; a fine line must be negotiated between an adequate resuscitation and one that is associated with the deleterious effects of fluid overload. Policies and practices have been highly individualized and can vary dramatically from institution to institution. However, the predominant teaching of the last quarter century has a pedigree derived from the influential publications involving regression analysis studies of resuscitative volumes in adult burn patients by Charles Baxter, at Parkland Hospital at Southwestern University Medical Center (Dallas, Texas, USA) in the 1960s. From these studies came the venerable *Parkland formula*,

which advocated the guideline for total volume of the first 24 hours of resuscitation (with *Ringer lactate* [RL] solution), at approximately 4 mL/kg body weight per % burn TBSA. With this formula, half the volume is given in the first 8 hours post-burn, with the remaining volume delivered over 16 hours.

Multiple formulas exist with variations in both the volumes per weight suggested and the type or types of crystalloid or crystalloid-colloid combinations administered (Table 1). To date, no single recommendation has been distinguished as the most successful approach. The RL solution is a relatively isotonic crystalloid solution that is the key component of almost all resuscitative strategies, at least for the first 24-48 hours. It is preferable to isotonic sodium chloride solution (i.e. *normal saline* [NS]) for large-volume resuscitations because its lower sodium concentration (130 mEq/L vs 154 mEq/L) and higher pH concentration (6.5 vs 5.0) are closer to physiologic levels. Another potential benefit of RL solution is the buffering effect of metabolized lactate on the associated metabolic acidosis. *Plasmalyte* is another crystalloid solution, the composition of which is even more closely physiological than RL solution, and Plasmalyte is used in some centers as the initial crystalloid solution for large burns. However, the significant cost difference per unit, with an uncertain benefit, has limited its widespread use at many burn units.

Regardless of the resuscitation formula or strategy used, the first 24-48 hours require frequent adjustments. Calculated volumes from all of the formulas should be viewed as educated guesses of the appropriate fluid load. Blind adherence to a derived number can lead to significant over-resuscitation or under-resuscitation if not interpreted within the clinical context. Over-resuscitation can be a major source of morbidity for burn patients and can result in increased pulmonary complications and escharotomies of the chest or extremities. In addition, not all burns require the use of the Parkland formula for resuscitation. Promptly addressed adult burns of <15-20% TBSA without inhalation injury are usually *not* enough to initiate the systemic inflammatory response, and these patients can be rehydrated successfully primarily via the oral route with modest IV fluid supplementation.

Table 1: Formulas used for resuscitation of burn patients

Formula	Fluid in 1 st 24 hours	Crystalloid in 2 nd 24 hours	Colloid in 2 nd 24 hours
Parkland	RL at 4 mL/kg/percentage burn	20-60% estimated plasma volume	Titrated to UOP of 30 mL/hour
Evans	NS at 1 mL/kg/ percentage burn, 2000 mL D5W, and colloid at 1 mL/kg/ percentage burn	50% of first 24 hour volume + 2000 mL D5W	50% of first 24-hour volume

Local Treatment of Burn Wounds

Knowing the mechanisms involved in wound healing is very important for effective treatment of burn wounds. The treatment strategy for the burn wound varies according to the extent and depth of injury.

Burn Wound Care at First Step and Emergency Department

Interventions that should be done at first step may differ according to the severity and mechanism of the burn. For *minor burns*, burned area should be put under running tap water for 20 min. Clothing that are soaked in hot liquid or contaminated with chemicals should be removed. For *chemical burns*, neutralizing agents should not be applied (neutralization reaction may cause further heat). Dry chemicals should be brushed away first and then irrigated with tap water. Before transfer to the designated facility, wounds should be wrapped with clean cloth but not covered with topical drugs. Topical silver sulfadiazine can be applied initially at the emergency department except for facial burns. Topical anesthetics are *not* recommended. Adherent dressings should *not* be used. Irrigation should be done with caution in order to avoid hypothermia due to cold water exposure. Depending on the severity of burn, wound care in a multidisciplinary burn center may be required.

Topics to be Covered in burn wound treatment

As previously mentioned, prolonged and exaggerated inflammatory response in deep burns results in intensified edema, which further delays wound repair and is associated with scarring. Although the anti-inflammatory treatments such as prostaglandin inhibitors and glucocorticoids carry the risk of impaired wound healing, it seems reasonable to diminish excess inflammation and edema in burn injury

Indeed, treatment with topical or low dose systemic glucocorticoids in the early phase of burns has been suggested to prevent aberrant inflammation. For *deep burns*, early excision and grafting are crucial to remove the foci of inflammation and infection.

Anti-inflammatory drugs including cytokine inhibitors, corticosteroids, interferon α and β , and methotrexate have also been used to prevent scar formation.

As the infection risk is increased in burn patients due to immunosuppression and the wounds can be rapidly contaminated by the organisms, prevention of infection should be the primary strategy in burn wound care. Disinfectants can be used without inhibiting wound healing, and wounds should be cleaned with tap water, saline, and non-irritant soaps. Early covering of burn wound with topical antimicrobial agents may prevent the invasion or contamination of the wound. In deep burns, micro-organisms may colonize the tissue below the eschar producing a source for infection. As the standard topical antimicrobials cannot penetrate the eschar tissue, early excision of the eschar is important in prevention of infection. Furthermore, early detection of infection is crucial especially in patients with deep and extensive burns. Prophylactic antibiotics are not recommended for burn wounds unless there is high probability of infection. In case of wound contamination and in immunocompromised patients (pediatric, perioperative, and diabetic patients), prophylactic antibiotics can be considered.

As the burn injury results in a profound hyper-metabolic state, nutritional support is recommended in order to enhance wound healing.

Burn Wound Coverage and Grafting

For the first-degree burn wounds

- Topical antibiotics are not necessary.
- Moisturizing agents are sufficient, and topical anesthetics may be given depending on the patient's condition.

For superficial second-degree burn wounds

- Although the burn wounds are sterile at the beginning of injury, the wound begins to be invaded by the organisms from the patient's flora or from the environment. Therefore, topical anti-microbials are recommended for superficial second-degree burn wounds.
- As silver sulfadiazine delays epithelialization, it can be used only for the first days to prevent infection.
- Wounds should be covered with non-adherent dressings including paraffin-impregnated gauze or ointments containing 0.2% nitrofurazone, zinc oxide, or dimethyl isopropylazulene. Several alternative topical agents have been suggested to be effective.
- Various types of dressing materials are available for the local care of burn wounds. Wound dressing selection should be tailored according to the amount of wound exudate, the presence of fibrin or necrotic tissue, and the depth of the wound. Hydrocolloids, hydrogels, chitin, polyurethane foams, alginates, and hydrofibers all have been recommended as treatment options for the local care of second-degree burn wounds.
- Blisters that may be seen in superficial second-degree burns may serve as an excellent environment for the growth of micro-organisms and increase the risk of infection. Small blisters may be left intact; however, large blisters should be removed, and the wound should be dressed.

For deep second-degree and third-degree burns

- The removal of the necrotic tissue, prevention of infection, and the maintenance of a moist environment are the primary goals to facilitate the wound healing in deep burns.
- Eschar is the tough, leathery necrotic tissue seen in full-thickness burns. Circumferential eschar tissue may compromise circulation on extremities or restrict breathing over the chest. Escharotomy may be indicated in these patients. In the case of compartment syndrome, fasciotomy should be performed. Eschar tissue does not break down spontaneously, except in the case of infection. Although the necrotic tissue of small deep burns may be treated by topical necrolytic agents, surgical debridement is needed in extensive burns. As spontaneous healing is not expected and the scar formation is the final outcome of deep second-degree and third-degree burns, early excision of the eschar and grafting are the preferred treatments for these wounds. After the excision of eschar, temporary wound covering for the first days by topical antimicrobials (silver sulfadiazine) or wound dressings prevents infection and maintains moisture before surgery.
- As the formation of scar can be prevented by the early and appropriate management of burn wound, excision should be done as soon as the patient is stabilized. Although the split-thickness auto-grafts are the gold standard method in deep burns, they have many

disadvantages. Allografts and xeno-grafts may serve a good option for larger burns until the allografts are incorporated; however, they have also many limitations. Tissue engineering has provided a new era in the wound care field. Skin tissue regeneration by tissue-engineered products showed promising results in wound healing. Tissue scaffolds, healing-promoting factors (growth factors), stem cells, and gene therapy are the current solutions provided by bioengineering.

- Although the experimental studies with either embryonic or adult stem cells demonstrate the potential use of stem cells in the treatment of chronic wounds, further research is required to investigate their long-term effects on wound healing process. Gene therapy is a promising approach for the future treatment of burn wounds. It involves the transfer of genes into cells that encode growth factors required for enhancing wound repair. However, its use in burn wounds is limited by technical challenges. In conclusion, further trials are required to explore the long-term effects and safety of tissue engineering methods in burn wound treatment.

Other Treatment Modalities

The common goal of all therapeutic tools above-mentioned is to optimize wound healing, prevent scar formation, and minimize the functional disability. There are more other treatments that have been used for these purposes with varying success.

- *Hyperbaric oxygen therapy* has been suggested as a safe and effective treatment for burn wounds and can be used in conjunction with other modalities for burn patients.
- *Silicone gels* have been suggested to be useful in burns which carry high risk of hypertrophic scarring. They are recommended to be used before the maturation of scar.
- *Platelet-rich plasma*: Experimental studies showed promising results in the wound healing with platelet-rich plasma (PRP) treatment; however, its routine use in burn wounds and scars requires further evaluation.
- *Laser*: Various types of lasers including pulse dye laser (PDL) and fractional ablative laser may offer better results when used in combination with surgery.
- *Pressure garments and massage therapy* are also used to minimize scar contraction.
- *Burn rehabilitation, splintage, and physiotherapy* are very important to prevent contractures and to improve functional outcome.
- *Psychological treatment*: Additionally, as burn survivors may experience significant psychosocial problems, proper specialists should be consulted as soon as possible

COMPLICATIONS OF BURN INJURIES

Burns cause both systemic and local complications. The major factors contributing to systemic complications are breakdown of skin integrity and fluid loss. The most common systemic complications are hypovolemia and infection. Local complications include eschars, contractures and scarring.

Systemic Burn Complications

The greater the percentage of total body surface area (TBSA) involved, the greater the risk of developing systemic complications. Risk factors of severe systemic complications and mortality include all of the following:

- Second- and third-degree burns of $\geq 40\%$ of TBSA
- Age > 60 years or < 2 years
- Presence of simultaneous major trauma or smoke inhalation

Hypovolemia

Hypovolemia causing hypoperfusion of burned tissue and sometimes shock, can result from fluid losses due to burns that are deep or that involve large parts of the body surface; whole-body edema from escape of intra-vascular volume into the interstitium and cells also develops. Also, insensible fluid losses can be significant. Hypoperfusion of burned tissue also may result from direct damage to blood vessels or from vasoconstriction secondary to hypovolemia.

Infection

Infection, even in small burns, is a common cause of sepsis and mortality, as well as local complications. Impaired host defenses and devitalized tissue enhance bacterial invasion and growth. The most common pathogens are streptococci and staphylococci during the first few days and gram-negative bacteria after 5 to 7 days; however, flora are almost always mixed.

Metabolic Abnormalities

Metabolic abnormalities may include hypo-albuminemia that is partly due to hemodilution (secondary to replacement fluids) and partly due to protein loss into the extravascular space through damaged capillaries. Dilutional electrolyte deficiencies can develop; they include hypomagnesemia, hypophosphatemia, and hypokalemia. Metabolic acidosis may result from shock. Rhabdomyolysis or hemolysis can result from deep thermal or electrical burns of muscle or from muscle ischemia due to constricting eschars. Rhabdomyolysis causing myo-globinuria or hemolysis causing hemo-globinuria can lead to acute tubular necrosis and acute kidney injury.

Hypothermia

Hypothermia may result from large volumes of cool intravenous fluids and extensive exposure of body surfaces to a cool emergency department environment, particularly in patients with extensive burns.

Renal and Electrolyte Complications

These include acute renal failure (early or late), acute adrenal infarction and derangement of metabolism and electrolytes (hypoglycemia, hypernatremia, hypokalemia, etc).

Respiratory Complications

Respiratory complications rank as the major cause of death in burn patients. Potentially fatal respiratory complications include inhalation injuries, aspiration of fluids by unconscious patients, bacterial pneumonia, pulmonary edema, obstruction of pulmonary arteries, and post-injury respiratory failure

Gastrointestinal Complications

Gastrointestinal complications are not uncommon after severe burns. *Ileus* is common after extensive burns. *Curling ulcers* may also occur.

Psychological and Social Problems

Burn survivors may develop severe psychological and social problems affecting their quality of life.

Multiple Organ Failure (MOF)

Despite recent advances, multiple organ failure (such as cardiac instability, respiratory or renal failure) and compromised immune function, which results in increased susceptibility to subsequent sepsis, remain major causes of burn morbidity and mortality

Local Burn Complications

Eschar Formation

Eschar is stiff, dead tissue caused by deep burns. A circumferential eschar, which completely encircles a limb (or sometimes the neck or torso) is potentially constricting. A constricting eschar limits tissue expansion in response to edema; instead, tissue pressure increases, eventually causing local ischemia. The ischemia threatens viability of limbs and digits distal to the eschar, and an eschar around the neck or thorax can compromise ventilation.

Scarring and Contractures

Scarring and contractures result from healing of deep burns. Depending on the extent of the scar, contracture deformities can appear at the joints. If the burn is located near joints (particularly in the hands), in the feet, or in the perineum, function can be severely impaired. Infection can increase scarring. *Keloids* form in some patients with burns, especially in patients with darker skin.

Malignancy on Burn Scars

Marjolin's ulcer is a rare cutaneous malignancy which may develop in burn scar. It occurs approximately two to three decades after the burn and is commonly seen on lower extremities as verrucous lesions. Squamous cell carcinoma is the most common form. The prevention of scar carcinoma by the early and effective treatment of scar formation is of primary importance to reduce the associated morbidity and mortality. Additionally, regular follow-up of patients with burn scars and early detection and evaluation of the non-healing ulcers are important considerations

CAUSES OF DEATH

Causes of death from burn injuries include the following:

1. Hypovolemia (refractory and uncontrolled) and shock
2. Renal failure
3. Pulmonary edema and ARDS
4. Septicemia
5. Multiple organ failure (MOF)
6. Airway block in head and neck burns

Vascular Anomalies

CLASSIFICATION

Vascular Anomalies are swellings with an abnormal abundance of blood vessels. They are classified into two main subtypes; vascular tumors (hemangiomas) and vascular malformations. Infantile hemangiomas (IHs) are benign vascular neoplasms characterized by excessive proliferation of vascular endothelial cells. In contrast, vascular malformations are structural anomalies resulting from abnormal development of vascular channels during embryogenesis (in-born errors of vascular morphogenesis). Malformations with an arterial component are rheologically fast-flow; the remainder is slow-flow.

Vascular Tumors

Benign

- Infantile hemangioma
- Congenital hemangioma
- Rapidly Involuting
- Non-involuting
- Partially Involuting
- Pyogenic granuloma (reactive hyperplasia rather than a true neoplasm)
- Others

Locally Aggressive / Malignant

- Kaposiform hemangio-endothelioma
- Tufted angioma

Vascular Malformations

- Capillary
- Venous
- Lymphatic
- Arterial/Arterio-venous
- Combined (complex/mixed malformation such as capillary-lymphatic-venous malformation)

INFANTILE HEMANGIOMA (IH)

Epidemiology

- It is the most common tumor of childhood
- Up to 10% of Caucasian/white children by year 1 - 1% in African-American children
- Female: Male ratio is 3:1
- Predominance for head and neck region (60%)

Risk Factors

- White race
- Premature infants / Low birth weight – up to 30%
- Maternal factors – advanced maternal age, multiple gestation pregnancy, placenta previa, and preeclampsia.
- Other risk factors: in-utero diagnostic procedures (chorionic villus sampling and amniocentesis), use of fertility drugs or erythropoietin, breech presentation, and being first born.

Anatomic Configuration (Figure 1)

- *Localized* (focal): discrete single lesion.
- *Segmental*: larger lesions, usually plaque-like, covering regions probably determined by neuroectodermal placodes. Segmental IH has higher risk of associated congenital anomalies
- *Multifocal*: multiple focal lesions, 5 or more lesions may be a marker for hepatic IH.
- *Indeterminate*: cannot definitively be categorized as localized or segmental



Figure 1: Different forms (anatomical configuration) of infantile hemangioma

Natural Progression

At Birth

- No identifiable lesion at all or signs of the incipient hemangioma, including an erythematous macule/spot, a telangiectatic mark, a faded area, “Herald” patch is pale macule, with central telangiectasia.

Proliferative Phase

- Initial rapid growth for the first few months of life, usually is sustained to the end of the 1st year but rarely persists beyond.

Involution

- Can begin as early as 6 months of age
 1. Color fades- appear dusker, purple, or gray. Gray-white areas appear from center
 2. Gradual decrease in thickness and volume
 3. Lesion less tender, softer
 4. Later - telangiectasia, atrophic wrinkled pale skin may result, dyschromia, or frank scarring 2^{ry} to ulceration.
- Involution rate
 1. By 5 years, 50% involuted
 2. By 7 years, 75% involuted
 3. By 9 years, 90% involuted
- Involution results
 1. Best results when involuted by age 4 years.
 2. Excellent cosmetic result = “no redundant skin, scar, or telangiectasia” in almost 50% of cases or more.
 3. Or after involution, 20-50% retain:
 - Residual bulk of fibro fatty tissue
 - Skin atrophy
 - Hypopigmentation
 - Residual telangiectasias
 - Scarring
 - Ulceration.

Complications

1. Ulceration/Infection (local, sepsis) (Figure 2)
 - Breakdown of the IH skin surface (tense rapidly proliferating). Trouble sites (anogenital region > upper lip > neck, chest)
 - It is the most common complication with an estimated incidence of 5% to 21%.
 - Ulceration can lead to significant pain, bleeding, and secondary infection and Scarring.



Figure 2:
Ulceration
(breakdown of
skin
breakdown)

2. Involvement of vital structures:
 - Obstructive airway (sub-glottic IH)
 - Eye (visual impairment): IHs occurring within the orbit have the potential to cause mechanical ptosis, strabismus, or astigmatism, which can quickly lead to the development of amblyopia.
 - Peri-oral region
 - Feeding impairment
 - Perineal/perianal region
3. Body image / disfigurement /destructive – facial features
4. Cardiac failure (liver lesions, and large lesions anywhere)
5. Special patterns (e.g. segmental)/ associated congenital anomalies

Imaging

- Imaging is usually *not* necessary.
- Ultrasound is generally the preferred modality for diagnosis, whereas MRI is better to assess extent of the lesion.
- Imaging may be required in the following cases:
 1. Diagnosis is uncertain
 2. Evaluation of extent is necessary
 3. When IH is a possible marker of other congenital disorders – syndromes
 4. Response to therapy needs to be monitored.

Management

Uncomplicated IHs

- Generally, do not require medical or surgical intervention till complete involution (*watchful waiting*)
- Possible future management of post-involution residual tumor.

Complicated IHs

- *Appropriate emergency* treatment of potentially life-threatening complications such as severe hemorrhage and acute airway obstruction.
- *Appropriate urgent* treatment of existing or imminent functional impairment, pain, or bleeding
- *Evaluation* to identify structural anomalies potentially associated with IH
- *Elective treatment* to reduce the likelihood of long-term or permanent disfigurement. Options include medical, surgical and laser therapy.

1. Oral Propranolol

- Now 1st line of treatment when indicated. Most of cases start to involute rapidly after initiation of treatment
- Often continued until at least 8-12 months of age
- Pre-treatment
 - Exclusion -bronchospasm, cardiac disease, CNS vascular anomalies
 - Baseline measurements/labs (BG), +/-ECG, +/-echo

- Dosing
 - Initial dose: 0.5 mg/kg/d, divided (2-3 times/day)
 - Increase to 2 mg/kg/d divided (2-3 times/day) over 2 days to 3 weeks as outpatient
 - Monitor
 - Initially in hospital for 1st 3 doses –BP, HR, glucose and temperature 1 h after dose
 - Observe for bronchospasm
 - Hold dose for HR <100, low temp, BG or BP
 - Parents observe for signs of lethargy, poor feeding, bronchospasm
 - Common SE: sleep disturbances, night terrors
2. *Topical Timolol cream* could be used for superficial lesions.
3. *Systemic Corticosteroids*
- It is no longer considered first-line therapy for IH due to its adverse effects.
 - 2nd line therapy. in case of contraindication or failure of propranolol (2-3 mg/kg/day)
4. *Intra-lesional steroids*
- May be used cautiously in well-circumscribed, small IH.
 - Adverse effects
 - Central retinal artery occlusion (periorbital lesions)
 - Eyelid necrosis (periorbital lesions)
 - Atrophy
 - Hematoma
 - Skin depigmentation
 - Possible significant systemic delivery
5. *Surgery*
- Indications of surgery for IH during *infancy* are limited:
 - Failure or contraindication of medical treatment
 - Focal involvement in an area favorable for resection.
 - A high likelihood that resection will ultimately be necessary, and the scar will be the same regardless of timing.
 - During /after involution, surgery may be indicated for
 - Excision of residual fibrofatty tissue,
 - Resection of scarred/lax skin and/or reconstruction of damaged structures.
 - Timing
 - Reasonable at 4 years. Self-esteem and long-term memory begin to form and the tumor has completed most of its involution.
6. *Laser treatment of IHs*
- May be useful in:
 - Treatment of ulcerating lesions; “multimodal” therapy.
 - Management of persisting post involution telangiectasia.
 - Ablation of early, non-proliferating, superficial lesions ?
 - Improvement of any post involution scarring.
 - Pulsed dye laser (PDL) is used most commonly.
 - Use of laser on proliferating and superficial IHs may lead to ulceration.
 - Atrophic scarring and hypopigmentation are also potential complications.

VASCULAR MALFORMATIONS (VMs)

Definition

An error in morphogenesis of any combination of the following vascular channels: arterial, venous, capillary, and lymphatic.

Unlike Hemangiomas

- VMs are present at birth (although may be noticed by the parents at later age)
- They grow proportionally to the size of the child
- No spontaneous involution.

Capillary Malformations / Port-Wine Stains (Figure 3)

- Most common type of VMs
- malformations of capillaries and venules in the upper dermis
- Histological- normal-appearing ectatic capillaries
- 0.3% newborns
- Frequently on face, CN V distribution.
- Over time they thicken, turn from red to purple, and from smooth to nodular.
- PSL is the main line of treatment.



Figure 3: Capillary Malformations / Port-Wine Stains

Venous Malformations (Figure 4)

- Can arise from any tissue including bone
- Small blue spongy blebs to diffuse venous ectasia
- Light to dark blue
- Empty by light compression
- No thrill/bruit.
- Painless on palpation unless thrombosis

- Swell when dependent
- Painful episodes related to thrombosis
- Skeletal hypertrophy



Figure 4: Venous malformations

- Treatment
 1. Supportive: Compression of extremity lesions
 2. Interventional Radiology: Sclerotherapy - Coil ablation
 3. Surgical Excision - debulking

Arterio-venous Malformations (Figure 5)

- Area of discoloration – red to purple
- Mass develops underneath
- Localized warmth to touch
- Associated thrill or murmur
- Tissue/limb hypertrophy
- Frequent late diagnosis - Not infrequently diagnosed later in childhood or adolescence



Figure 5: Arterio-venous malformation

- Treatment (frustrating)
 1. Complete surgical excision is the goal
 2. Difficult to control bleeding during surgery. Pre-operative embolization is valuable.
 3. Recurrences common with incomplete excision.
 4. Ligation of feeding vessels only makes lesion worse?!

Lymphatic Malformations (Figure 6)

- Previously known as “cystic hygroma” or “lymphangioma”
- Based on vessel size classified into - Macrocytic / microcytic
- Clear cutaneous vesicles may be seen – skin involvement– “Tip of the iceberg”
- Often found in the neck and intraoral
- Does not respect anatomic boundaries - Can infiltrate organs
- Supple mass - Soft and compressible
- May change in size -Wax and wane with infections
- Trans-illuminate
- No bruit or warmth



Figure 6: Lymphatic malformations

- Treatment
 1. Supportive: Compression of extremity lesions
 2. Interventional Radiology: Sclerotherapy for macrocystic variant.
 3. Surgical Excision: difficult

VASCULAR MALFORMATIONS: TREATMENT SUMMERY

Supportive measures can be helpful

CM -- PDL laser

VM -- Sclerotherapy or surgical excision

LM -- Sclerotherapy (macro-) or surgical excision

AVM -- Combined embolization / surgical excision

Congenital Anomalies of the Lip and Palate

EMBRYOLOGY

- The face develops from 5 processes; a *fronto-nasal*, 2 *maxillary* and 2 *mandibular* processes (Figure 1). Each is formed of mesoderm, a covering of ectoderm and a lining of endoderm. Failure of development and fusion of those processes → different forms of facial cleft.

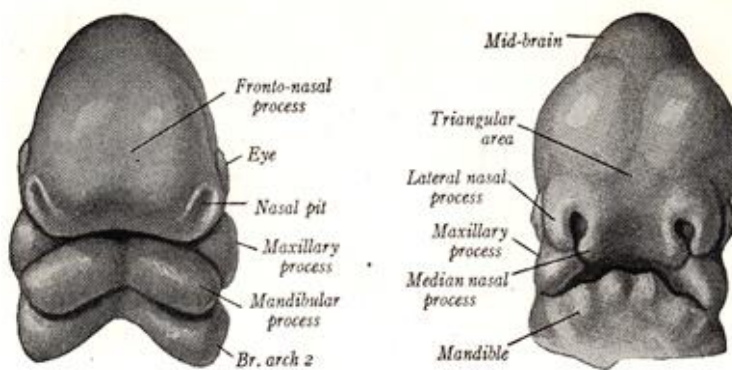


Figure 1. Embryology of the face

- The maxillary processes develop into the lateral segments of the upper lip, while the fronto-nasal process forms the central portion of the lip (prolabium), the nose and premaxilla. These elements are collectively known as the *primary palate* (structures anterior to incisive foramen).
- The soft and hard palates develop from two lateral segments of the maxillary processes (called palatal shelves) that are vertically oriented in early embryo. At 12-weeks, they fuse together from anterior to posterior after being rotated upwards. Failure in development & fusion of those palatal shelves results in different forms of palatal clefts. All structures posterior to incisive foramen are known as the *secondary palate*. Figure 2 shows anatomy of the palate

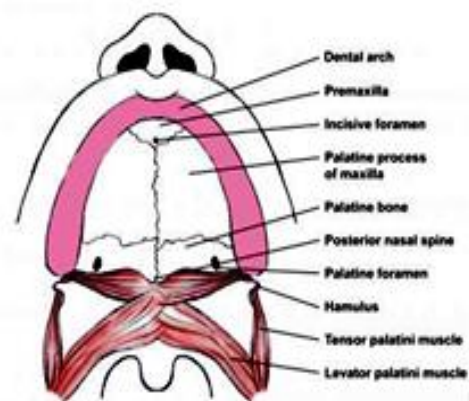


Figure 2. Anatomy of the palate

PATHO-PHYSIOLOGY

- By virtue of the multiple actions of the orbicularis oris muscle, the lips are responsible for competence of the mouth and actions of feeding (grasping food, sucking liquids), whistling, blowing, kissing and for production of certain phonemes (P, B, V, W, O, U) during articulation.
- Similarly, the levator palati and tensor palati muscles of the soft palate are responsible for conditioned separation of the oro-pharynx from the naso-pharynx during acts of chewing, deglutition and sneezing, and are responsible for the differential passage of air through the mouth and nasal cavities during articulation. Regurgitation of food to the naso-pharynx, where the Eustachian tubes are, can → otitis media and hearing loss in cleft palate patients, and the defective deglutition mechanism can → to chest infection due to food aspiration
- The hard palate is responsible for separation of the mouth cavity from the nasal cavity and creation of the negative pressure needed for suckling.
- The alveolar ridge of the maxilla (the dental arch) is responsible for dentition, and the teeth that can be affected in cases of cleft lip and palate are the lateral incisors, canines, and first premolars. Those teeth can be absent, supernumerary, or ectopic.
- The nasal anatomy is closely related to the lip and palate anatomy, therefore the nasal deformity is an integral component of the cleft lip & palate deformity.

ETIOLOGY

The exact cause of cleft lip and palate remains unclear and is thought to be multi-factorial:

1. Hereditary (genetic aberration).
2. Non-hereditary (first trimester exposure to teratogenic agents).

INCIDENCE

- *Race*: Inter-racial differences exist in the incidence of cleft lip and palate. The mean incidence of cleft lip and palate is 2.1 cases per 1000 live births among Asians, 1 case per 1000 live births among white people, and 0.41 cases per 1000 live births among black people.
- *Type*: 2/3 of cases involve the lip ± palate and 1/3 involves isolated cleft palate.
- *Gender*: Cleft lip is more common in males, while cleft palate is more common in females.
- *Side*: Unilateral clefts are 4 times the bilateral clefts, with more predilection to the left side.
- *Associated anomalies*: Approximately, 5% of cases of cleft lip & palate are associated with identifiable syndromes of craniofacial anomalies.

CLASSIFICATION AND CLINICAL TYPES

Cleft lip and palate can be broadly categorized into 3 groups (Figure 3):

- Group I Clefts anterior to the incisive foramen.
- Group II Complete cleft lip and palate.
- Group III Clefts posterior to the incisive foramen.
- In each group, different forms and degrees of severity exist, e.g. cleft lip can be unilateral or bilateral, simple or complete (involving the nostril floor), and cleft palate can be just a bifid uvula, a cleft soft palate, or a cleft hard & soft palate. Group II clefts (complete cleft lip and palate) can be unilateral or bilateral.

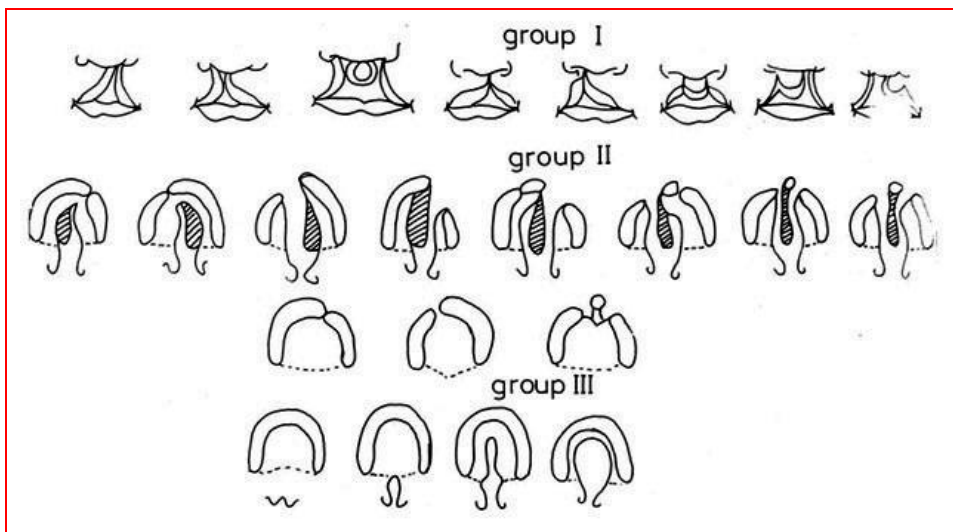


Figure 3. Groups and forms of cleft lip and palate

- In complete cleft lip and palate, the 2 cleft segments are totally separated from each other and are therefore subject to mechanical forces that exaggerate the cleft deformity with time. Thus, these cases need pre-operative alignment of the two cleft segments by orthodontic appliances.
- Figures 4-9 depict various clinical forms of cleft lip and palate.



Figure 4. Unilateral simple cleft lip



Figure 5. Bilateral simple cleft lip



Figure 6. Unilateral complete cleft lip



Figure 7. Unilateral complete cleft lip and palate



Figure 8. Bilateral complete cleft lip and palate



Figure 9. Cleft hard and soft palate

MANAGEMENT

Cleft lip and palate deformity requires multidisciplinary management involving the cooperation of plastic and maxillofacial surgeon, orthodontist, pediatrician, ENT specialist, speech pathologist, and pediatric psychologist in order to optimize the treatment outcome.

Pre-operative Procedures

- *Newborn care:* accurate diagnosis of the deformity and any associated conditions such as airway maintenance (in cases associated with retrognathia or *Pierre Rubin* anomaly) and feeding rehabilitation (naso-gastric feeding if needed, drops or spoon feeding, special teats for bottle-feeding, palatal obturators and upright feeding position to prevent nasal regurgitation).
- *Parent consultation* regarding magnitude of the condition and plan of management.
- *Careful monitoring* and support of weight gain, middle ear, chest and general condition.
- *Naso-Alveolar Molding (NAM)* by orthodontic appliances, in case of complete cleft lip and palate, to approximate the gap in the alveolar ridge, to expand the shortened columella and to reform the deformed nostril, in order to optimize the conditions for a successful surgical repair (Figure 10). Naso-Alveolar Molding also provides a palatal obturator that facilitates feeding.



Figure 10. Pre-operative Naso-Alveolar Molding to align the alveolar ridges and reform the nostril shape

Surgical Repair

- *Aim of surgery:* Adequate restoration of the functional anatomy of orbicularis oris muscle, nasal muscles, palatal muscles and palatal mucosa, and aesthetic reconstruction of the nostrils, lip skin and mucosa.
- *Time of surgery:* early repair of cleft lip and palate avoids exaggeration of the deformity and is needed to restore the balance of facial growth and the function of the palate before the age of articulation (15-18 months).
 - Isolated cleft lip: 2-6 months (average 3 months).
 - Complete cleft lip and palate: first stage surgery at 2-6 months (average 3 months) to repair the lip, nostrils, alveolar ridge and anterior part of hard palate – second stage surgery at 9-15 months (average 12 months) to repair the remaining part of the hard palate and the soft palate.
 - Isolated cleft palate: 9-12 months.
- Figure 11 demonstrates the immediate result of surgical repair in case of unilateral complete cleft lip and palate, and Figure 13 demonstrates immediate result of surgical repair in a case of bilateral complete cleft lip and palate.



Figure 11. Surgical repair of unilateral complete cleft lip and palate



Figure 12. Surgical repair of bilateral complete cleft lip & palate.

Follow-up and Secondary Procedures

- Post-operative nasal molding to improve nostril shape.
- Continued care of nutritional status and middle ear condition.
- Speech rehabilitation and training.
- Orthodontic management of malocclusion and aberrant teeth; including the possible need for alveolar bone grafting at the age of 2^{ry} dentition (7-9 years).
- Surgical management of complications (closure of oro-nasal or palatal fistula, revision of lip scar) or residual deformities (nostril asymmetry, nasal septum deviation, broad nose).
- Velo-pharyngeal surgery (to correct functional incompetence of the soft palate).
- Psychological and social support.

Hand Injuries

ANATOMY OF THE HAND

Bony Anatomy

- The **wrist** is composed of 8 *carpal bones* arranged in 2 rows of 4 bones each. The flexor *retinaculum* together with the carpal bones forms the *carpal tunnel*.
- The *metacarpal bones* articulate with the wrist at the carpo-metacarpal (CMC) joints.
- The *thumb* has only one inter-phalangeal (IP) joint, while the rest of the digits have proximal inter-phalangeal (PIP) & distal inter-phalangeal (DIP) joints.

Intrinsic Muscles of the Hand

They can be divided into 4 groups as follows:

1. The *thenar eminence* is formed by the extensor pollicis brevis and 3 short thenar muscles; abductor pollicis brevis, flexor pollicis brevis and opponens pollicis. Innervated by the recurrent branch of median nerve. The superficial location of this branch renders it vulnerable to trivial trauma to the thenar eminence. The adductor pollicis is innervated by ulnar nerve.
2. The *hypothenar eminence* is formed by a group of 3 muscles of the palm that control the motion of the little finger; abductor digiti minimi, flexor digiti minimi.
3. *Lumbricals*: They flex the digits at the MCP joints and extend the IP joints. They place the fingers in the writing position.
4. *Interosseous muscles* (n=7) are located between the metacarpal bones; 3 are palmar and 4 are dorsal. The palmar interossei adduct, while the dorsal interossei abduct the fingers.

CLASSIFICATION OF HAND INJURIES

According to the Type of Injury

1. *Incised* wound (treated by 1st sutures).
2. *Crush* injury, either open (treatment with 2nd closure), or closed (requires a decompression incision).

According to Cleanliness of the Wound

1. *Tidy*: Clean, neat lacerations e.g. sharp knives and glass injuries.
2. *Untidy*: Potential contamination, tissue devitalization and foreign body (FB) implantation. Ragged, saturated, uneven wound e.g. industrial injuries, saw injuries.
3. *Indeterminable*: Severe crush injuries and burns.

According to the Structure Injured

1. Skin or soft tissues.
2. Finger-tip injuries.
3. Vascular injuries.
4. Tendon injuries (flexors and extensors).
5. Nerve injuries.
6. Fractures and joint injuries.
7. Amputations.

PRINCIPLES OF MANAGEMENT

Early Care

Protect the recently injured hand against infection and damage.

1. Sterile dressing with immobilization in *position of function*.
2. Cleaning the surrounding skin with water & soap.
3. Anti-tetanic serum (ATS) and gas gangrene toxoid.
4. Antibiotic drugs in full doses (IV).
5. Snug dressings and elevation in case of severe bleeding.
6. Wound treatment is started when suitable conditions are available i.e. adequate anesthesia, surgically aseptic technique, proper instruments, good light, and a bloodless operative field.
7. Immobilization of all major hand injuries is a must.

Good History-Taking

1. Time of accident.
2. Place of accident.
3. Causative agent.
4. Mechanism of injury.
5. Type of first aid treatment and medications received.

Evaluation

1. Infection status of the wound: Clean, grossly contaminated, infected, etc.
2. General nature of wound: Lacerated, crushed, etc.
3. Source of major bleeding.

4. Test for damage assessment of the injury sustained):
 - Is there any tissue loss? e.g. bone, tendon, muscle, ect ?
 - What structures are exposed ?
 - What is the viability of the skin ?
 - What structures are damaged ?
 - Tendon injury: Assessed by finger movements. Check for range of motion (ROM) at rest and against resistance.
 - Nerve injury: Assessed by sensation, *not* motor power, which has been already impaired.
 - Bone injury (X-ray) and joint stability (only under general anesthesia).
 - Foreign bodies (FBs) are detected by X-ray.
5. Assessment of functional loss.
6. Assessment of function requirements of the patient

Definitive Treatment of Hand Injuries

1. Thorough cleansing of the entire hand & forearm while protecting the wound should be performed (brush with soap & water → scrub, cleanse and trim finger nails).
2. Gentle cleansing of the wound with soap and water, then scrubbing the wound and forearm with Betadine solution.
3. Inspection of the wound with adequate exposure and additional incisions in natural creases are done if needed.
4. Use of tourniquet for hemostasis is extremely important.
5. Wound debridement: Removal of all FB, excision of devitalized tissues and irrigation of the wound with Ringer's Lactate.
6. Repair of soft tissue structures in clean wounds of short duration but never of wounds with established infection.
7. Choose the best suitable method of treatment for the patient according to:
 - *Hand dominance.*
 - *Texture of the skin.*
 - *Patient's occupation* (typist, pianist, etc).
 - *Specific digital movement*, specially the thumb (50% of the hand), index and middle finger.
 - *Age* (e.g. cross finger flap in the elderly should be left for about 4 weeks, to avoid stiffness in both fingers).
8. Repair of different injured tissues as follows: Injuries of bone, nerves, tendons, muscle, skin, nail-bed and finger tips.

BONE INJURIES

Diagnosis

- Careful palpation
- X-ray examination.
- CT-scan may be required.

Treatment

- Closed reduction and splint if possible.
- Open reduction with K-wire internal fixation is indicated in:
 1. Fractures involving a joint surface.
 2. Metacarpal and proximal phalangeal fracture causing shortening, rotation, or mal-alignment.
 3. Multiple fractures.
- After treatment:
 1. Maintain the hand in the *position of function* (not ease): i.e. flexion of metacarpophalangeal (MP) joints, extension of inter-phalangeal (IP) joints, abduction of thumb and dorsi-flexion of wrist.
 2. Elevation of the hand.
 3. Early physiotherapy.

NERVE INJURIES

Types of Repair

1. Primary repair
 2. Delayed repair
- Primary repair of severed nerve ends of the digital nerve, median nerve, or ulnar nerve, using magnifying loupes or the microscope is the *best line of treatment* (Figure 1)

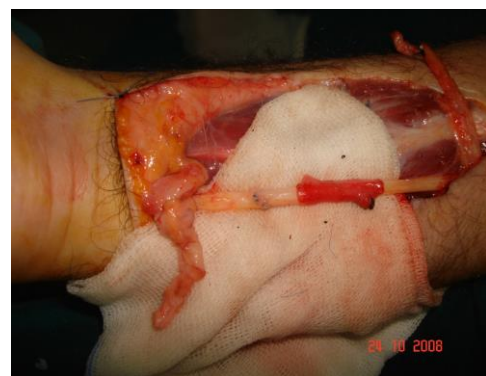
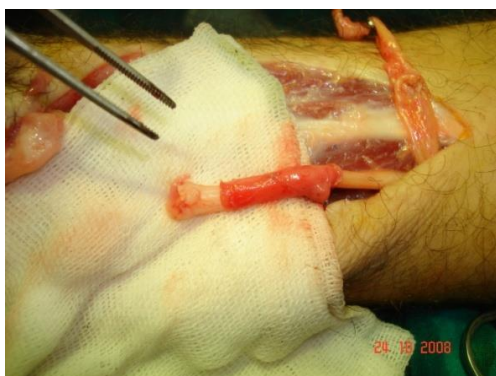


Figure 1. Primary repair of nerve injury (method of choice)

- Nerve repair is done either by the epi-neural or peri-neural technique.
- Nerve grafts as 1st treatment should be performed by an experienced hand surgeon.

TENDON INJURIES

Anatomy of the Tendons

Anatomy of the Flexor Tendons

- *Flexor digitorum superficialis* (FDS) tendons maintain constant arrangement in the distal wrist: The tendons to the middle and ring fingers lie palmar to those of the index and little fingers.
- *Flexor digitorum profundus* (FDP) tendons travel in a single layer deep to the superficialis tendons in the wrist & the palm.
- The *lumbrical muscles* originate from the FDP distal to the carpal tunnel.
- Over the proximal phalanx, the FDS tendon splits into 2 slips around the FDP tendon & then reunites deep to it with decussation of half of the fibers (*Camper's chiasma*) (Figure 2).

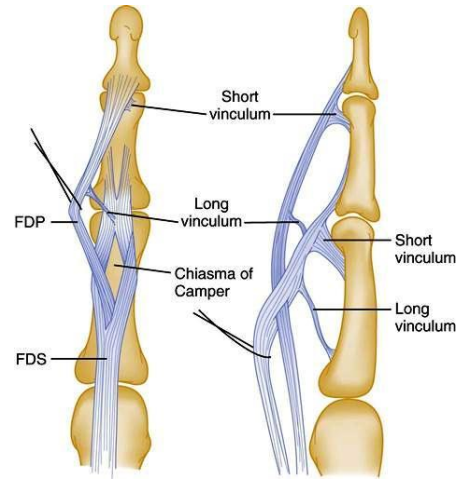


Figure 2. Tendons of FDS and FDP.
Note Camper's chiasma

- Pulleys of the fingers (Figure 3) consist of a palmar aponeurosis pulley, 5 annular pulleys (A) and 3 cruciate pulleys (C) (Figure 4). The annular pulleys A2 and A4 are crucial for normal digital function, they prevent tendon bowstringing and provide optimal joint flexion (Figure 5)

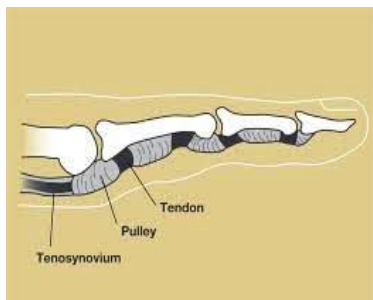


Figure 3. Pulleys of the flexors of fingers

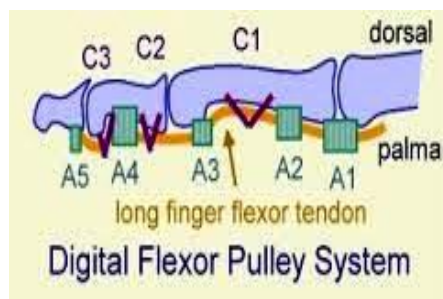


Figure 4. "5" annular pulleys (A) and "3" cruciate pulleys (C)

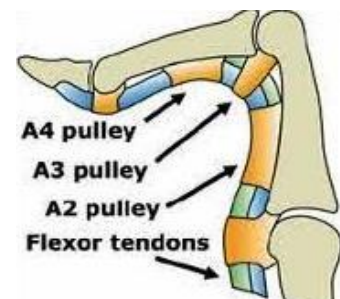


Figure 5. The crucial pulleys (A2 & A4)

Principles of Tendon Repair

Diagnosis

- Tendon injuries can be diagnosed by examination of the range of motion or movement (ROM) at the wrist, MP and IP Joints both at rest and against resistance (Figure 6).

Optimal Conditions of Repair

- Tendons heal rapidly when held in *apposition* and strong union occurs at 4 weeks.
- Tendon repair should therefore be performed under optimal conditions, which include the following: (1) Adequate anesthesia, (2) complete aseptic technique, (3) bloodless field (tourniquet), (4) good magnification, (5) atraumatic technique and (6) adequate exposure.



Figure 6. Clinical diagnosis of flexor tendon injuries.

Timing and Type of Repair

- 1st repair Within 24 h (*1st repair is mandatory*)
- Delayed 1st repair ... 1-14 days
- Early 2nd repair 2-5 weeks
- Late 2nd repair > 5 weeks
- Tendon grafts.
- Tenolysis

Suture Techniques of Repair

- Any suture technique should include:
 1. Core suture.
 2. Circumferential epitenon suture.
- Many suture techniques for flexor tendon repairs have been advocated; **Bunnell, modified Kessler-Tajima, augmented Becker, Strickland, Kleinert, Pulvertaft and epitenon-first.**
- Both ends of the cut tendon are delivered into the wound. If the proximal end has been retracted, the end can usually be palpated beneath the skin and a small accessory incision can be made over it. The end is withdrawn into the wound with fine artery forceps.
- The ends are trimmed and sutured with a **modified Kessler stitch** (grasping stitch) or **modified Bunnell stitch** (criss-cross stitch) using a non- absorbable suture material (Figure 7).
- If the tendon is avulsed from terminal phalanx, it is re-attached, tying the suture over the nail.
- The tourniquet is released and skin sutured

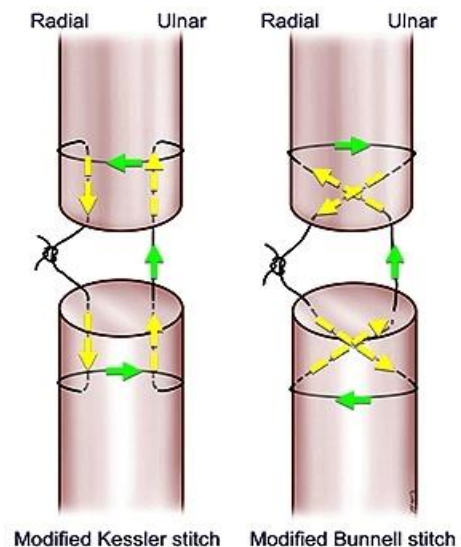


Figure 7. Technique of tendon repair

- *After care*
 1. A gauze dressing is applied to the wound and the hand and fingers are bandaged over a pad of puffed-up gauze in the palm with the wrist and fingers flexed.
 2. A plaster back slab is added from the elbow to the distal IP joint.
 3. The hand is elevated for 48 hours.
 4. Gentle movements of the digits within the dressings are allowed.
 5. Plaster and dressings are removed after 3 weeks. Fingers and hand are mobilized under supervision.

Post-operative Mobilization

- Tendons become adherent to surrounding tissues easily, which limits their gliding movement. Such adhesions can be limited by mobilization. There are 3 methods of post-operative motion.
- The method used should be tailored according to the patient as follows:
 1. *Non-compliant patient* → Controlled passive motion (**Duran & Houser**) (Figure 8).
 2. *Compliant patient* → Controlled active extension (**Kleinert**) (Figure 9)
 3. The fingertip (nail) is attached by a rubber band to the plaster splint of the forearm; because of rubber band recoil, the patient can *actively extend his finger, with a passive flexion*.
 4. *Highly motivated patient* → Early active motion (**Chow**)
 5. This is carried out under strict supervision of a physiatrist then by the patient. The aim is to do selective 5 daily active FDP and FDS flexion, separately. This will help the differential function of the separate muscles.

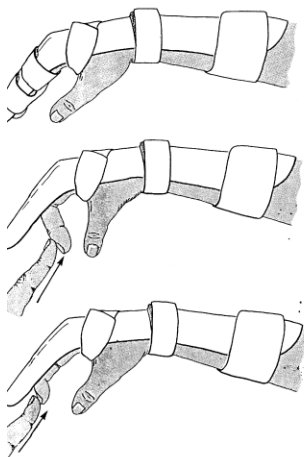


Figure 8.
Duran and Houser controlled passive motion



Figure 9. Kleinert's elastic of the flexor tendon of the middle finger

- The results of immediate repair of extensor tendons are better than those of flexor tendons (due to the presence of the flexor sheath in the latter).

Flexor Tendon Injuries

- **The actual level of tendon injury** in relation to its surrounding tissue is of significance in estimating the prognosis. According to the site of injury, they are classified into 5 zones (**Verdan's flexor tendon zones**) (Figure 10).

- Zone I → From insertion to proximal IP joint
- Zone II → From proximal IP joint to distal palmar crease
- Zone III → Mid-palmar space
- Zone IV → Wrist
- Zone V → Forearm

• **Methods of retrieval of flexor tendons:**

There are multiple maneuvers to retrieve the flexor tendons for repair::

1. Blind retrieval.
2. Catheter retrieval.
3. Tendon retrievers.
4. Endoscopy.

- **Flexor tendons are sutured** with 1^{ry} repair, or 2^{ry} late repair using tendon graft from the palmaris longus or plantaris.

• **Tendon injuries on the palmar aspect of the hand**

- **Zone A (Distal to the proximal IP joint):** The divided flexor digitorum profundus (FGP) is repaired primarily (if the condition permits), otherwise a flexor tendon graft via the terminal phalanx with a pullout wire is performed.
- **Zone B (No man's land of Bunnell = the danger area) (from proximal IP joint to distal palmar crease):** Traditionally, if both flexor tendons are divided (superficialis and profundus), close the skin alone and at a 2nd operation resect the flexor digitorum superficialis (FDS) & repair the FDP by a tendon graft (because adhesions from 1^{ry} repair at this level would lead to stiffness and poor function). If only the FDP is divided, disregard lest repair endangers the FDS. However, in expert hands, good results are obtained after 1^{ry} repair of both with Klinert's post-operative rubber band traction.
- **Zone C (from the distal palmar crease to the wrist):** Both FDP & FDS are repaired by 1^{ry} sutures (if the condition permits).

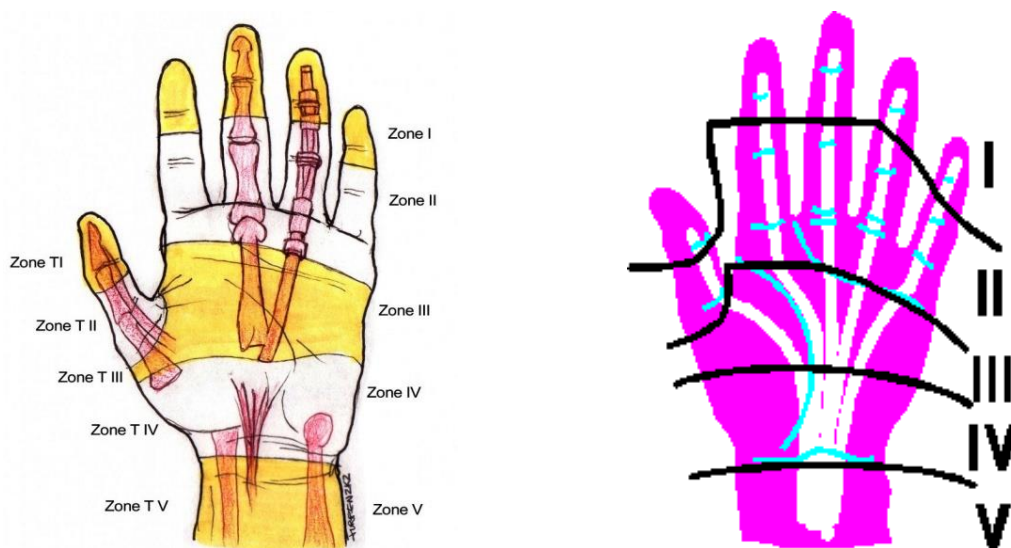


Figure 10. The 5 zones of the flexor surface of the hand and fingers

Extensor Tendon Injuries

Primary suture is the treatment of choice and presents no particular problems apart from the Mallet finger and Boutonniere's deformity mentioned below. The injury could be at the following sites:

1. *Near the insertion:* This → “Mallet deformity” due to flexion of the terminal phalanx. This insertional injury may be associated with avulsion fracture of the base of the phalanx. Treatment is by repair of the tendon (+ splint for 6 weeks, if there is avulsion).
2. *At the middle of the phalanx:* The middle slip of the extensor tendon will be damaged, while the 2 lateral slips will be dislocated forwards around the sides of the joint & come to act as flexors resulting in “Boutonniere's deformity” with flexion of proximal IP joints & extension of distal IP joints. Treatment is by splinting the proximal IP joint in extension for 6 w using a splint is successful, if started early. Cases that present late may be left untreated if the disability and deformity are slight. Otherwise an attempt must be made to repair the central slip or reconstruct it by using one of the lateral bands.
3. *At the dorsum of the hand (level of MP joints):* The capsule and the MP joints are very superficial at this site & so will be opened. Therefore, they should be repaired before repair of the tendons, otherwise the result will be a stiff joint and a rough surface on which the tendon should glide. Tendon repair is done by “wearing” (i.e. capsule → soft tissue → tendon → skin).
4. *At the dorsum of the hand (one or more):* Its repair is easy as no joints or nerves are present. The dorsal carpal ligament may be divided, or even excised, to allow fine sliding of the tendons.
5. *At the dorsum of the lower forearm or at the wrist:* Tendons are repaired primarily & muscles are sutured (otherwise the gap will be filled with fibrous tissue).

Tendon Grafting

- Staged procedures, in which a new mesothelial “tube” forms around an implanted silicon rod and a graft is subsequently threaded through the pseudo-sheath, have become well established in the treatment of difficult injuries, or failed repairs, where scarring prevents early reconstruction.

Tendon Transfer

- Superficialis (FDS) to flexor pollicis
- Transfer sets in radial and posterior interosseous nerve injuries.
- Opponens plasty

Repair of Muscle Injuries

- Muscle injuries should be repaired (sutured) after good debridement of all necrotic fibers

SKIN INJURIES

- Debridement of skin laceration or untidy wounds may result in skin defect, which may or may not be accompanied by exposure of deep structures.
- **Defects with exposure of important deeper structures:** This needs flap coverage. Skin flap may be *local* (eg. Kutler V-Y random; volar V-Y flap, dorsal transposition flap), *regional* (cross finger, palmer), *distant* (groin; abdominal or from the chest) or *free flap*.
- **Defects without exposure of deeper structures:** This is covered by free skin graft.

NAIL BED INJURIES

Type of Injuries

- Cut wounds
- Lacerations
- Complete crushing.

Treatment

- The nail should be preserved & the nail bed should be reconstructed by fine suture material.
- Sub-ungual hematoma should be evacuated if >25% of the size of the nail either by trephination or removal of the nail plate

FINGER TIP INJURIES

Definition

- The tip of the finger means “the part of the finger which lies between the insertion of the profundus tendon (distal end) and the extreme of the finger

Type of Injury

1. Incised or cut wound (e.g. home accidents such as knife injuries).
2. Blunt or crush injuries (e.g. industrial or door injuries).

Assessment of Injury

- Injuries *without* bone exposure
- Injuries *with* bone exposure.

Management

- *Injuries with no bone exposure*
 1. A transverse or slightly oblique *small* (< 1cm) wound in the fingertip can be treated by healing with 2^{ry} intention, using only mild antiseptic solutions (especially in children).
 2. In larger (*moderate*) wounds, the split skin graft (SSG) is done, especially in adults as epithelization is not as rapid as in children.

- *Injuries with bone exposure:*

If the distal phalanx is exposed, *no* SSG is used because exposed bone does *not* “take” the graft, besides, it will ulcerate due to trauma. The type of closure depends on the size of the wound. Usually a *flap* will be needed (local, regional, distal or free).

1. *Cross-Finger Flap*

- The full-thickness injury is adequately debrided, and skin of the dorsal aspect of the adjacent finger is elevated to the lateral midaxial line to preserve the blood supply and is sutured to the palmar aspect of the injured finger. The fingers are immobilized in mild flexed position, and a SSG is used to cover the donor defect (Figure 11). The flap is released in 2-3 weeks.
- A thick, soft, viable, cornified, more or less sensitive skin is supplied by this flap, and the patient can leave the Hospital in just one hour.



Figure 11. Cross-finger flap of a full-thickness finger-tip injury

2. *Advancement V-Y Flap*

- A V-shaped incision is made proximal to the well-debrided injured area.
- The triangular area contained in the V-shaped incision is advanced forwards to cover the injured area. The incision is then closed (Y-shaped) as seen in Figure 12.

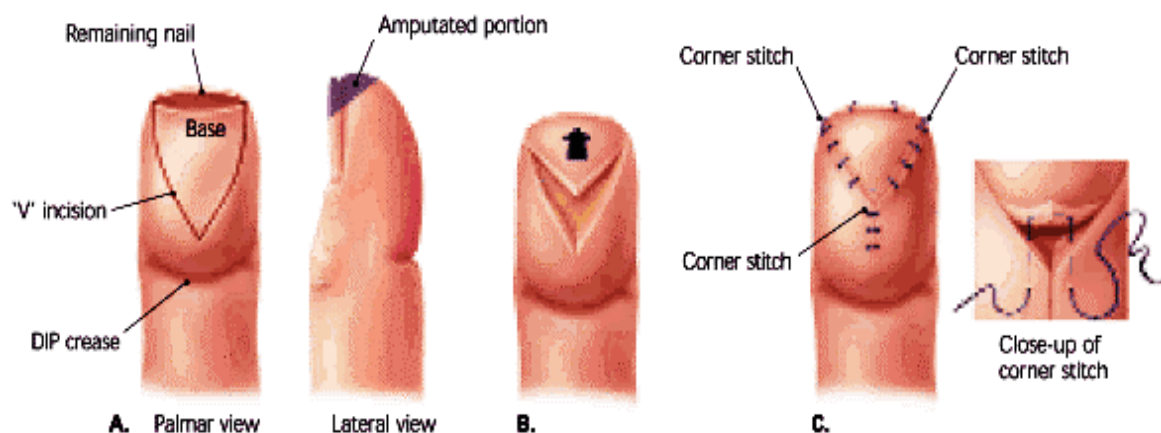


Figure 12. Advancement V-Y flap for coverage of a full-thickness finger-tip injury

3. *Dorsal transposition of skin to the tip (terminalization)*

- It is used in higher amputations with available dorsal skin that can be transposed to cover the injured area.

AMPUTATION INJURIES

Re-implantation

Definition

- Replantation is the reattachment of a completely detached body part.
- Fingers and thumbs are the most common but the hands, ear, scalp, arm and penis have all been replanted.

Microsurgery of the Hand and Fingers

- Generally replantation involves restoring blood flow, restoring the bony skeleton and connecting tendons and nerves as required.
- The use of magnifying microscopes, fine instruments, very fine suture materials 8-0 to 11-0 have made it possible to replant amputated fingers and hands successfully.

Requirements

- Good preservation of amputated parts (in a bag containing crushed ice).
- Quick transfer to hospital.
- Preservation of the hand of the patient by applying dressing **only**. No trial should be attempted to stop the bleeding by clamps.

Indications and Contraindications of Re-implantation

- Absolute indications: Amputated thumb, multiple fingers, or hand, in children and young age.
- Contraindications: severe crush, old age, marked instability.

Outcome of Re-implantation

- Initially, success was defined in terms of a *survival* of the amputated part alone. However, as more experience was gained in this field, surgeons began to understand that survival of the amputated piece was not enough.
- Thus, *functional demands* of the amputated specimen became paramount in guiding which amputated pieces should & should not be replanted.

CUT WRIST (FLEXOR ASPECT)

1. Quick assessment, with fluid replacement if still bleeding from cut arteries.
2. *Vascular injury*: Immediate *repair* of at least one artery (radial and/or ulnar) with 6/0 Prolene.
3. *Nerve injury (usually median nerve)*: Immediate *1^{ry} repair* should be always attempted.
4. *Tendon injury*: Immediate *1^{ry} tendon repair* should always be performed as mentioned above.
5. *The Skin*: Skin closure is essential, as raw areas, predispose to infection and fibrosis.
6. *Antibiotics & anti-tetanus toxoid* should be administered routinely.

Hand Infections

GENERAL PRINCIPLES

Introduction

- Until the advent of antibiotics, infections of the hand often resulted in severe disabilities, including stiffness, contracture and amputation.
- Improper treatment and delay in instituting appropriate therapy, can lead to disastrous outcome.
- High dose of systemic antibiotics and proper splinting within 24-48 hours can arrest the condition.

Etiology

- *Causative organisms:* Staphylococci (80%), streptococci and Gram +ve bacilli (20%)
- *Predisposing Factors:*
 1. History of trauma (50%).
 2. Lymphatic or blood spread.
 3. Manual workers & house wives who frequently suffer from abrasions & pricks.
 4. Site (limited blood supply or easily choked off e.g. synovial sheath, bone, joints, nail folds).

Clinical Picture

Complaints

1. Marked pain and toxemia (particularly with deep suppuration).
2. Severe tenderness.
3. Extensive edema involving the area affected, but it should be noted that pitting edema of the dorsum of the hand is usually due to deep infection on the palmar side of the hand.
4. Movements of related fingers and joints become painful and limited.
5. Deep suppuration is suspected by hectic fever, throbbing pain interfering with sleep and increasing when the hand is dependent and marked leukocytosis.

History-Taking

1. The mechanism or the process by which infection began.
2. Its immediate functional effects on the hand.
3. The patient's general health e.g. DM, alcoholism, smoking (all delay resolution of infection).
4. Drug addiction & method of injection.

Physical Examination (should proceed in the following sequence):

1. Hand inspection, in comparison to the other hand.
2. Circumferences are measured, comparing both sides, for objective follow-up of resolution.
3. Areas of cellulitis are circumscribed with a pen to provide objective baseline information.
4. Evidence of proximal spread is sought by careful search for red streaks & palpable LNs.
5. Visible draining pus is sent for culture and sensitivity (C&S).
6. Search for other areas of infection in drug addicts.
7. The patient is asked to put the part(s) through an active range of motion and limitation is noted.
8. Passive motion is gently checked with attention to tendon and joint pain.
9. Parts are gently palpated for fluctuation.

Diagnostic Studies

1. *Plain X-Ray*: To check skeletal involvement, gas in soft tissues, or retained FBs.
2. *Ultrasound* is helpful for evaluation of subtle sheath infection and dead space infection.
3. *Bone scan* is useful to screen for distant foci. It is sensitive, but not specific
4. *MRI* is excellent for early osteomyelitis and marrow edema. It is very sensitive, but not specific.

General Measures of Treatment (5 Principles)

1. *Rest and elevation*:
 - *Rest* of the hand (splinting) in the *semiflexed position* of all joints (position of function). It should be applied loosely to avoid venous and lymphatic outflow obstruction.
 - *Elevation* of the hand to ↓ edema and pain.
2. *Antibiotics* in full doses.
3. *Early recognition of pus*.
4. *Drainage* of pus, usually under general anesthesia.
5. *Adequate aftercare*
 - Active exercises
 - Rehabilitation).

Classification

Type	Causes
Generalized Infections	- Lymphangitis - Thrombophlebitis - Cellulitis.
Localized Infections 1. Skin and SC infections 2. Infection of fascial spaces 3. Infection of synovial sheaths 4. Infection of Bones and Joints	Subcuticular abscess - SC abscess - Paronychia - Distal pulp infection (Felon) – Superficial web space infection. Palmar sub-aponeurotic space infection - Midpalmar space infection (deep palmar abscess) - Thenar space infection - Infection of the forearm (Parona's space infection) - Dorsal (SC & subaponeurotic) space infection. Suppurative tenosynovitis - Ulnar bursitis - Radial bursitis. - Osteomyelitis - Septic arthritis.
Other Infections	- Bacterial: Tuberculosis (TB) & Leprosy. - Fungal: Sporotrichosis & other fungal infections. - Viral: herpes simplex (HS), Orf. - Human bite – Animal bite. - Barbers pilonidal sinus (PNS). - Rare infections – Necrotizing fasciitis

GENERAL INFECTIONS**Lymphangitis****Etiology and Clinical Picture**

- Organisms (sterptococcus) gain entrance via abrasions, which may be microscopic.
- Superficial lymphangitis signifies proximal spread of streptococci from a distal infection. Within few hours, the hand becomes swollen, painful and warm, and then red streaks appear coursing up the arm.

Sites of Affected LNs

- Lesions of the ulnar half of the hand → epitrochlear LNs.
- Infections of the midfinger → LNs above the clavicle (deltopectoral LNs).
- Infections of the thumb & index → axillary LNs.

Treatment

- *Medical:* Rest and elevation, antibiotics, analgesics, antipyretics + treatment of the primary condition.
- *Surgical:* Only when there is abscess of axillary LNs → incision and drainage (I&D).

Superficial Suppurative Thrombophlebitis

It is a serious infection because of ↑ tendency among drug addicts to inject impure substances.

Drainage

- Drug injection infection that does *not* respond to IV antibiotics within 12-24 hours, are explored surgically. The vein, SC space, fascia and subfascial regions are evaluated.
- When the pus exudes from the aspirated or incised lumen of a vein during surgery, the vessel is dissected out well proximally through a separate incision and ligated. The distal end is then ligated and the vein is stripped with an external vein stripper.
- After vein excision, a pediatric catheter with multiple perforations is secured into the vein bed for irrigations every 2 h with saline or antibiotic solution for 1 to 2 days.

Cellulitis



Figure 1. Cellulitis of left hand

Clinical Picture

- It involves only the skin and is characterized by erythema, warmth, edema, pain of localized area with glossy, tight appearance of the skin. It *must* be documented that deeper structures are *not* involved with painless full (range of motion (ROM) of digits, hand and wrist.
- There is no tenderness on palpation of deeper structures. The commonest organisms are Strept. pyogens and occasionally Staph. aureus

Treatment

- Immobilization and elevation
- Antibiotics e.g. Augmentin

LOCALISED INFECTIONS

(A) SKIN and SC INFECTIONS

Subcuticular Abscess

- *Anatomy:* Infection under the cutis, usually under a callosity.
- *Treatment:* Excision of the cutis over the abscess.

Subcutaneous Abscess

- *Anatomy:* Infection under the dermis due to neglected puncture wounds or infected hematomas.
- *Clinical Picture:* A red, tense, tender, swelling. Soft edema spreads around the affected area, but *not* to the dorsum and this differentiates it from deep infections.
- *Treatment:* *Drainage* through an incision over the most fluctuant area parallel to the nearest flexion crease, under proximal regional nerve block anesthesia. Collar button abscesses are drained through separate palmar & dorsal incisions connected by a penrose drain through the wound. A small ellipse of skin is excised as the palmar incision is made, to prevent premature closure of the wound. Drainage wounds are packed open with Vaseline impregnated gauze. The hand is remobilized within 2-3 day when edema subsides.

Paronychia / Eponychia

- *Anatomy:* Infection of tissues around and deep to the nail.
- *The causative organism* is Staph. aureus. In case of thumb sucking/nail biting, the organisms are anaerobes. If chronic, the causative organism is candida (fungal).
- *Clinical Picture:* It is the *most common infection in the hand*, caused by frequent trauma of this area. The organism often enters through a hangnail, causing painful throbbing reddish swelling in the skin edge at one side of the nail (nail fold) (Figure 2) that may extend to the other side (collar-shaped). It may have associated cellulitis. If it extends to the overlying proximal nail, it is called *eponychia*. Abscess formation is possible.
- *Treatment*
 1. Early cases: Warm soaks & elevation.
 2. *Drainage* under digital block anesthesia. Soften by soaking .A number 11 blade is used. If infection is severe with purulent collection beneath the nail, part of it should be removed. Follow up 24-48 hours. Most resolve in 5-10 days. Antibiotics alone are rarely effective.
 3. Late cases: Marsupialization and partial nail removal + antibiotics ± antifungal.



Figure 2. Paronychia. Note redness and edema of the nail fold & the appearance of pus.

- *Complications*
 1. Osteomyelitis of the distal phalanx
 2. Eponychia (subungual abscess)
 3. Felon formation
 4. Chronic infection (most likely fungal).

Distal Pulp Infection (Felon)

- *Anatomy*: Infection of the pulp of the distal finger or thumb closed space. This pocket (distal pulp) is subdivided into 16 compartments by fibrous septa, which run in the long axis of the phalanx & are attached to both the skin & periostium (Figure 3). Although these septa facilitate infection & inhibit drainage, yet they act as a barrier protecting the joint space & tendon sheath. The distal pulp is isolated by the distal finger crease, which limits proximal spread.
- *Etiology*: Penetrating trauma and 2^{ry} infection by Staph. Aureus.
- *Clinical Picture*: a severely painful, red, tender, hard swelling due to high tension (Figure 4), but fluctuation is late when necrosis has occurred.
- *Treatment*:
 - Early treatment: warm soaks and anti-staph/strept antibiotics.
 - Late treatment:
 - Surgical drainage under digital block via (1) longitudinal midline palmar surface incision (go directly to the pus), (2) lateral vertical incision just in front of the bone if it results from extension of paronychia, (3) in severe cases, 2 lateral incisions are made to transfix the space dividing all septa.
 - If there is osteomyelitis, bone debridement is done and pus is sent for C&S + oral antibiotics for 5-7 days. The wound is irrigated, packed open and the finger elevated. Re-evaluate in 24-48 hours and re-pack if necessary.
- *Complications*: Osteomyelitis (requiring partial finger amputation), necrosis of palmar surface and sinus tract formation, suppurative flexor tenosynovitis, shaft necrosis, and septic arthritis.

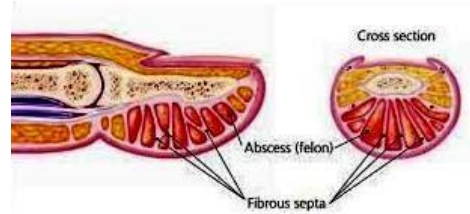


Figure 3. Fibrous septa of distal pulp.



Figure 4. Felon (distal pulp infection)

Superficial Web Space Infection

- *Anatomy*: The web space extends from the distal flexion crease of the palm to the bases of the fingers. There are 3 web spaces between the 4 slips of palmar aponeurosis. These contain the interossei, lumbricals, digital vessels & nerves. The web space of the thumb that contains the adductor pollicis & 1st lumbrical muscle. Infection of this space occurs 2^{ry} to infection of palmar blisters.
- *Sequelae*: Pus tracks between the digital process of the palmar fascia to form a bilocular (collar-button) abscess, one loculus under the skin & the other under the palmar fascia. Pus may also track along the related lumbrical canal to reach the deep spaces of the hand

(midpalmar or thenar spaces), but usually the abscess points in the web before this can occur.

- *Clinical Picture:* Red painful throbbing swelling under the callosed skin and in the web. Localized edema and tenderness, separation of the related fingers and dorsal edema of the hand.
- *Treatment:* Transverse incision is made on the palmar aspect over the affected web space parallel to the distal flexion crease, over the point of maximum tenderness.

(B) INFECTION OF FASCIAL SPACES

Palmar Sub-aponeurotic Space (Subfacial Web Space)

- *Anatomy:* It lies between the palmar aponeurosis and flexor tendons. It is the 1st space where infection occurs and is usually connected to a SC infection forming a *collar-stud (collar-button) abscess*.
- *Clinical Picture:* the abscess obliterates the concavity of the palm with *no* dorsal edema.
- *Treatment:* Small transverse incision over the SC infection + widening of the opening in the palmar aponeurosis to drain the deeper collection.

Midpalmar Space Infection (Deep Palmar Abscess)

- *Anatomy:* It is a deep, serious abscess posterior to the flexor tendons and anterior to the interossei and metacarpals. *Medially*, a flexor membrane separates it from the hypothenar eminence and *laterally* a strong fibrous septum separates it from the thenar eminence (Figure 5).

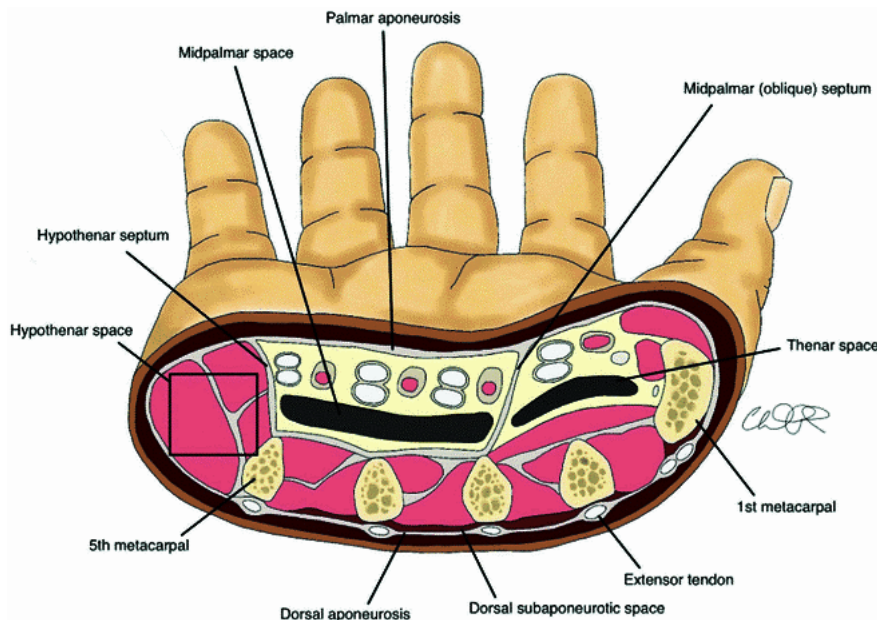


Figure 5.
Anatomy of the facial spaces of the hand.

- *Etiology:* It usually results from (1) *spread of infection* from web spaces or from the tendon sheath of the little, ring and middle fingers, from the ulnar bursa or the thenar eminence, and rarely by lymphatic spread from superficial infections, and (2) direct penetrating injury.

- *Clinical Picture:* Marked swelling of the whole hand (loss of hand concavity) with marked edema of the dorsum (frog hand) + Pain with movement of 3rd and 4th digits and severe tenderness of the central palm. General examination reveals fever, tachycardia, previous operations, exhaustion by pain & insomnia.
- *Treatment:* (1) *Transverse palmar approach:* incision along the middle 1/3 of the distal flexion crease. Deepen the drainage without injury of tendons or nerves or arteries (2) *If pus extends well proximally*, the carpal tunnel is released through a separate incision to prevent destruction of the median nerve by pus under pressure. An irrigation catheter is passed through the carpal tunnel & the overlying skin is loosely re-approximated with 2 sutures to avoid desiccation of the nerve.

Thenar Space Infection

- *Anatomy:* It is bounded *anteriorly* by the palmar fascia and flexor tendon of the index and thumb, *posteriorly* by the adductor policis, *medially* by the flexor septum of the thumb and *laterally* by the shaft of the 1st metacarpal.
- *Etiology:* (1) Penetrating injury into thenar space, (2) SC abscess of the thumb or index fingers, (3) tenosynovitis of the thumb or index finger, or (4) extension (direct spread) from the radial bursa or mid-palmar space.
- *Clinical Picture:* History of a 1^{ty} infection. Pain and swelling (ballooning) of the thenar eminence and 1st web space + marked edema of the dorsum of the hand, but the palm retains its concavity. The thumb is in abduction and flexion.
- *Treatment:* (1) incision along the lateral border of the dorsum of the 2nd metacarpal dividing the 1st interosseus muscle, or (2) a curved incision along the lower border of the 1st dorsal interosseus muscle, an artery forceps is introduced along the anterior surface of the adductor policis. In addition, an irrigation catheter is inserted in the space.
- *Complication:* (1) Functional disability, (2) sepsis, (3) tendon destruction and (4) hand loss.

Infection of the Forearm (Parona's Space)

- *Anatomy:* This space lies in the distal part of the forearm between the pronator quadratus and interosseus membrane *posteriorly* and the flexor tendons, radial and ulnar bursae *anteriorly*, and at each side by deep fascia which blends with periosteum and SC margins of the ulna and radius.
- *Etiology:* Usually occurs as a spread from midpalmar space infections, ulnar or radial bursae.
- *Clinical Picture:* (1) pain and edema in the lower part of the forearm which shows brawny tender induration, (2) rigidity of the hand and forearm, the arm being held in a position of flexion at the wrist and elbow, (3) deteriorated general condition of the patient.
- *Treatment:* Incision along the medial border of the ulna. A counter-incision may be needed at the radial side. The contents of the carpal tunnel are carefully retracted toward the thenar eminence. An irrigation catheter is placed and situated posteriorly.

Dorsal Spaces Infections

- *Anatomy:* The dorsum of the hand contains 2 fascial spaces, the dorsal SC space and the dorsal subaponeurotic space, which lies between the extensor tendons and the dorsal interossei muscles and is closed on each side by fusion of the extensor aponeurosis with the periosteum of the 2nd and 5th metacarpals.
- *Clinical Picture:* History of a punctured or lacerated wounds or osteomyelitis of the metacarpals. Signs of local inflammation are present looking like cellulitis. Swelling and erythema on dorsum of the hand, pain with passive movement of extensor tendons and severe tenderness even before the presence of pus.
- *Treatment (Drainage):* (1) The SC space is drained by a vertical incision over the point of greatest tenderness. (2) The subaponeurotic space is drained by an incision along the ulnar border of the 5th MC or medial border of the 2nd.

(C) INFECTION OF SYNOVIAL SHEATHS

Anatomy of the Synovial Sheaths (Figure 6)

- The flexor tendon of each finger lies in a fibro-osseous tunnel bounded posteriorly by the metacarpal (MC) head, the proximal and middle phalanges and the capsule of the intervening joints, and anteriorly by the fibrous flexor sheath. The digital synovial sheath lines the tunnel and invests the tendons.
- The middle 3 fingers have separate synovial sheaths, which end at the level of the distal palmar crease.
- The synovial sheath of the flexor tendon of the thumb extends into the forearm as the radial bursa.
- The sheath of the little finger is continuous with the common flexor sheath which is present in the palm and extends behind the flexor retinaculum into the forearm.



Figure 6. Anatomy of the synovial sheaths

Suppurative Flexor Tenosynovitis

- *Etiology*
 1. Neglected puncture wounds to the proximal and middle phalanges that form abscesses that decompress into the flexor sheaths.
 2. Neglected dorsal SC finger infection that spreads along the lumbrical canals to the proximal aspect of the flexor sheath.
- *Pathogenesis:* If left untreated, sheath infections will destroy the tendons by disruption of their synovial diffusion and vascular perfusion. Pus in the thumb and index finger sheaths spread proximally into the thenar space, whereas pus in the long, ring and little finger sheaths spreads proximally into the midpalmar space.

- *Clinical Picture:* It involves the middle 3 fingers. Diagnosis is reached by **Kanavel's 4 cardinal signs**; (1) symmetrical (fusiform) enlargement of the whole finger, (2) a flexion attitude, (3) pain along the flexor tendon with passive extension and (4) tenderness and erythema over the course of the sheath.
- *Treatment:* 2 transverse incisions; one over the distal finger crease and another at the distal flexion palm crease. Pus is evacuated & the space is irrigated with a 5-Fr pediatric feeding catheter (Figure 7). Before discharge, the ROM is recorded objectively by measurements of the total active motion (TAM).
- *Complications*
 1. Involvement of the forearm
 2. Suppurative arthritis of a related joint
 3. Stiff finger
 4. Paralysis of the median nerve



Figure 7. Drainage and irrigation of septic tenosynovitis

Ulnar Bursitis

- *Etiology:* It occurs as a complication of infection of the little finger.
- *Clinical Picture:* Swelling of the palm, with preservation of its concavity and with edema of the whole hand specially the dorsum (due to lymphatic spread). Painful extension of the little finger and maximum tenderness between the transverse palmar crease (**Kanavel's sign**).
- *Treatment:* Incision along the *radial* margin of the hypothenar eminence with division of the palmar aponeurosis. If extension into the forearm has occurred, the upper cul-de-sac of the bursa is drained through another incision along the anterior surface of the ulna.

Radial Bursitis

- *Etiology:* it occurs as a complication of infection of the thumb.
- *Clinical Picture:* Edema and tenderness of the thumb and lateral side of the hand. Flexion deformity and painful extension.
- *Treatment:* Incision on the *medial* surface of the thenar eminence (along the ulnar border of the thumb). Pressure is made over the upper extension in the forearm, and if pus egresses into the wound, another incision is made at the wrist over the tendon of the FCR. The bursa is identified by a probe passed proximally from the 1st incision and then incised and irrigated.

(D) INFECTION OF BONES AND JOINTS

Suppurative Arthritis

- *Etiology:* From direct inoculation from penetrating trauma or adjacent bony or soft tissue infection, bite wounds, or as a complication of inadequate treatment of hand infections.
- *Sites:* It may affect any joint, most commonly the proximal inter-phalangeal (PIP) and metacarpo-phalangeal (MCP) joints.
- *Diagnosis*
 1. The joint is red, swollen, with localized tenderness (unlike flexor tenosynovitis). There may be an overlying puncture wound.
 2. Passive motion is restricted and very painful.
 3. It is diagnosed by arthrocentesis.
 4. Plain X-ray shows widening of joint space caused by pus under pressure, or narrowing of joint spaces caused by chondrolysis from a virulent mixture of organisms.
- *Treatment:* Antibiotics to cover *Staph. Aureus*. Drainage to prevent extensive articular cartilage destruction by lysozymal activity.

Osteomyelitis

- *Etiology:* It is most common with open fractures or soft tissue infections.
- *Clinical Picture and Diagnosis:* It manifests by fever, redness, swelling, warmth, tenderness and pseudo-paralysis (in kids). Plain film shows bony destruction or periosteal elevation
- *Treatment:* Antibiotics (broad spectrum) and surgical debridement.

OTHER HAND INFECTIONS

A. Bacterial Infections

Tuberculosis (TB)

- It is usually chronic, relatively painless and involves one hand only. Bones and joints may be infected but more commonly the tendon synovium is involved and becomes matted to the tendon.
- *Treatment:* *Partial sheath excision* for mycobacterial tenosynovitis, sparing the key pulleys and *synoviectomy* for wrist synovitis + anti-TB drugs.

Leprosy

- Leprous neuritis of the median and ulnar nerves → sensory and motor loss of the hand and increased susceptibility of injury of the hands due to anesthetic digits.
- *Treatment*
 1. Anti-leprosy drugs
 2. Reconstructive surgery
 3. Occupational training.

B. Fungal Infections

Sporotrichosis

- It is an indolent chronic infection that occurs in gardeners, generally beginning with a prick from a thorn. The causative organism is the fungus *sporothrix Schenckii*. It presents with a red raised lesion at the inoculation site followed by lymphangitis ± arteritis and synovitis.
- *Treatment*: Potassium iodide or amphotericin B.

Other Fungal Infections

- Mainly involves the nails due to *candida albicans*, *trichophyton*, etc.
- *Treatment*: Anti-fungal drugs eg. Amphotericin B + local Nystatin and removal of the nail for chronic intractable cases.

C. Viral Infections

Herpes Simples (HS) (Herpetic Whitlow)

- *Etiology*: Caused by direct inoculation through broken skin infection of the hand; in kids with herpetic gingivo-stomatitis & health care workers
- *Clinical Picture*: Pain, pruritis, swelling, vesicles that coalesce over 2 weeks with ulcer formation that has a hemorrhagic base. It may look like a felon, but *drainage is contraindicated!!* Take a careful history of a tender distal finger but *soft* pulp space.
- *Treatment*: It resolves spontaneously in 3-4 weeks. Prevent oral inoculation by covering with a dry dressing. Acyclovir only if immuno-compromised or with frequent infections.

Orf (Contagious Pustular Dermatitis of Sheep)

- *Etiology*: Viral disease of sheep transmitted to humans who handle infected animals (butchers).
- *Clinical Picture*: It presents with multiple skin lesions, often on the flexor aspect of forearm and a wart pustular lesion on the finger.
- *Treatment*: anti-viral drugs.

D. Human Bite

- Typically occurs with clenched fist injuries from punch to mouth.
- Human saliva has demonstrated 42+ species such as Strept, S, Aureus, E. corrodens (30%) and Bacteriodes (the most common anaerobe).
- If not seen within 24 h, it should be assumed as being "infected".
- The wound over the MCP joints should be considered intra-articular until proven otherwise to avoid potential consequences of untreated septic arthritis.
- *Treatment*: surgical extension of wound and explore I&D

E. Animal Bite

- It results from dogs > cats > rodents. Cat bites are worse due to needle like puncture.
- Can be very dangerous due to infection with many pathogens including Vincent's organisms. *Pasteurella multocida* (facultative anaerobe) is the most common. Staph, Strep and anaerobes are common.
- A common injury is not only a bite, but an incised wound over the knuckles with the joints usually penetrated.
- *Treatment*
 1. X-rays Under nerve block anesthesia; excision of the wound including the capsule of the joint if affected with a 2 mm rim of tissues around and leaving the wound open.
 2. Infected tendons are debrided along with the skin.
 3. Antibiotics (penicillin + 1st-generation cephalosporin) are given for 4-7 days.
 4. Splinting of the hand in a position of function for 1-2 days.
 5. Plain X-ray is obtained to look for retained FBs and skeletal disruption or infection.

F. Barbers Pilonidal Abscess (Sinus)

- Usually occurs in the webs between digits 3 and 4 of the left hand due to penetration of the cut: hairs, which have beveled extremities similar to the point of a hypodermic needle.
- If not inflamed, the lesion is marked by a small black dot with area of epithelial scales around.
- Recurrent attacks of subacute and acute inflammation in the sinus cause pain requiring excision.

G. Rare Hand Infections

- *Etiology*: Gas gangrene, syphilis (S), deep fungal infections (coccidioidomycosis, actinomycosis), tularemia, anthrax, yaws (framboesia), glanders disease.
- *Diagnosis*: History of exposure, chronicity & laboratory studies to identify the pathogen.
- *Treatment*: According to the cause (ie. specific treatment).

H. Necrotizing Fasciitis

- It is a life- and limb-threatening emergency that is most commonly seen in intravenous (IV) drug addicts.
- *Organisms*: Single pathogen (Hemolytic Strep) or polymicrobial (H. Strep, Staph and anaerobes).
- *Clinical Picture*: It presents with extreme pain, rapid advancement, cellulitis with poor margins, tense swollen skin. Ecchymosis and bullae appear with time followed by leukocytosis
- *Treatment*: Findings include liquefaction of fat and fibrinous necrotic tissue, thrombosis of vessels, foul smelling "dish water" pus. Muscle is often spared. Rapid Wide surgical debridement of involved tissues (the most important factor for recovery + broad spectrum antibiotics.
- *Prognosis*: Poor prognostic factors- >50 years, chronic illness, DM, truncal spread.

Skin Tumors and Lesions

BENIGN SKIN TUMORS

A. Skin Appendages

1. *Sebaceous gland* hyperplasia and adenoma.
2. *Sweat gland* tumors (uncommon in clinical practice) e.g. cylindroma (turban tumor).
3. *Hair follicle* tumors (uncommon) e.g. Trichoepithelioma.

B. Common Cysts

Epidermal Cyst (Sebaceous Cyst)

- *Clinical Presentation*
 - *Site*: It is common in the scalp, ears, retro-auricular region, face, scrotum and thorax
 - *Age*: It occurs in adulthood and middle age (rare before adolescence).
 - It is usually attached at some point to the overlying skin, which cannot be pinched out, but is freely movable over the underlying structures unless there is a fibrous reaction due to infection.
 - Sometimes, it is marked by a delicate pore (punctum which the opening of the occluded duct) (Figure 1) with or without a comedo.
 - Consistency is cystic or doughy. It may be *indented* due to its doughy sebaceous material inside. *Squeezing* may cause the sebaceous material to come out through the punctum.
- *Complications* include *infection* (abscess formation), *sebaceous horn*, *Cock's peculiar tumor* (ulcer on top of the sebaceous cyst simulating epithelioma), *malignant transformation* (into sebaceous adenocarcinoma), atrophy of the hair follicles and baldness of the scalp.
- *It should be differentiated from* other diseases of sebaceous glands e.g. sebaceous adenoma or adenocarcinoma & from dermoid cyst (Table 1).
- *Treatment*
 - If *infected*: Incision and drainage (I&D).
 - If *clean*: It is excised in-toto with its capsule to avoid recurrence.

Dermoid Cyst

- It may be *congenital* (inclusion or sequestration dermoid) or *acquired* (implantation dermoid).
- It contains epithelium and adenexal structures.
- It is of variable size in the *SC plane*. Thus, it is *not* attached to the overlying skin.
- *Site*: The sequestration type is typically found in the post-auricular region, pre-auricular (temporal) region, inner and outer canthus (Figure 2), and *midline* of the scalp, chin, sublingual, supra-sternal, pre-sternal and pilonidal
- The cyst is usually single, small (1-2 cm), ovoid or spherical, cystic or soft, with a smooth surface and well-defined borders. It may fluctuate but will *not* trans-illuminate (opaque). It is *not* pulsatile, compressible or reducible. The skin overlying is normal unless it is inflamed.
- It contains a pultaceous material & shows positive moulding sign.
- *Treatment*: Surgical excision.



Figure 1. Sebaceous cyst. Note the punctum



Figure 2. Dermoid cyst at the outer canthus (lateral eyebrow).

Table 1: Differences between dermoid cyst and sebaceous cyst

Criteria	Dermoid Cyst	Sebaceous Cyst
- <i>Tissue plane</i>	SC tissue	Skin (dermal)
- <i>Age</i>	Childhood and adolescence	Middle and old age
- <i>Site</i>	At lines of embryonal fusion	Anywhere especially scalp and face
- <i>Punctum</i>	-ve	+ve
- <i>Pinching skin</i>	+ve (SC location)	-ve (intra-dermal)
- <i>Consistency</i>	Cystic or soft	Cystic or doughy (indented)
- <i>Multiplicity</i>	Usually single	Commonly multiple
- <i>Extension</i>	Intra-cranial (if in the scalp)	-ve

C. Soft Tissue Tumors

Papilloma

- It consists of a central axis of connective tissue, blood vessels and lymphatics.
- The surface is covered by squamous epithelium and may be smooth or villous.
- *Clinical Examination:* A pedunculated swelling of normal skin color, but may be pigmented, occurring anywhere on the skin, but mostly on the neck, trunk, face and intertriginous zones. It is soft, solid and not fluctuant. The LNs are not enlarged and local nerves and vessels are normal.
- *Treatment:* Excision

Fibroma

- True fibroma is rare.
- Desmoid tumor of rectus sheath is the commonest site. Extra-abdominal desmoids of muscle sheaths & fascia may also occur.
- It is characterized by a high recurrence rate (RR).
- *keloids* may be considered as a type of fibroma. Differences between keloid and hypertrophic scar are summarized in Table 2.

Normal scar

- It should be thin, linear and contain the minimum of scar tissue. However, sometimes, the fibrous tissue (FT) response is excessive and results in *hypertrophic scar*, or *keloid formation*.

Hypertrophic scar

- The excessive FT is confined to the scar.
- It is usually thick and red and may itch.
- It never gets worse after 6 months and does *not* recur after excision.

Keloid (In Greek = crab's claw):

- *Definition:* The hypertrophy and overgrowth of FT extends beyond the original wound into normal tissues. It follows wounds, burns, vaccination marks; but may occur spontaneously.
- *Factors Affecting its Occurrence:*
 1. Site: It is common on the face, neck, ears, over the sternum and front of abdomen.
 2. Lines of skin tension (scars crossing skin creases).
 3. Race: It is more common in Negroes.
 4. Age: It is less common in infants & in old people.
 5. Pregnancy: Incidence increases with pregnancy.
 6. Tuberculosis: It occurs more in patients with TB.

- *Characteristic clinical features* (Figure 3)
 1. It is raised above the surface of the skin.
 2. Pale pink in color, and firm in consistency.
 3. The surface may be lobulated or even claw-like.
 4. Margins are ill defined and project irregularly into the SC and sub-fascial tissues.
 5. It continues to get worse even after years.
 6. It becomes very unsightly, tender & usually itchy.
 7. Recurrence is common.
 8. It never becomes malignant.



Figure 3. Keloid left arm

- *Treatment:*
 1. Low-voltage X-ray therapy.
 2. In long-standing cases: Excision and re-suturing preceded and followed by radiation.
 3. Intra-dermal injection of steroids as shown in Figure 4.
 4. Shaving the keloid with resurfacing the area by a thin split graft.



Figure 4. Intra-dermal keloid injection

Table 2: Differences between keloid and hypertrophic scar

Features	Hypertrophic scar	Keloid
- Genetic	Not familial	May be familial
- Race	Not race related	Black > White
- Gender	Female = Male	Female > Male
- Age	Children	10-30 y
- Borders	Remains within the border	Outgrows wound area
- Natural history	Subsides with time	Rarely subsides
- Site	Flexor surface	Sternum, shoulder, face
- Etiology	Related to tension	Unknown !!

Lipoma

- It is a slowly growing tumor composed of fat cells of adult type (Figure 5), and may be encapsulated (Figure 6) or diffuse.
- *Types* according to site: e.g. Subfascial, intramuscular, submuscular, etc.
- It may contain other tissues e.g. fibrolipoma, angiolipoma, nevolipoma, etc.



Figure 5. Lipoma of the left shoulder



Figure 6. A capsulated lipoma, completely excised.

- It is characterized by a definite slippery edge and a lobulated surface. A sense of fluctuation may be obtained. Most lipomas are painless, but may be multiple & painful (neurolipomatosis).
- **Dercum's disease** (*adiposis dolorosa*) is characterized by tender deposits of fat especially on the trunk & arms (Figure 7).



Figure 7. Multiple lipomata in the left arm

- **Complications:** Large lipomas of the thigh, shoulder and retro-peritoneum occasionally undergo sarcomatous changes. A lipoma may also undergo myxomatous degeneration, saponification and calcification.
- **Treatment** is excision and insertion of a drain if there is a large cavity left behind.

Neuroma

- True neuromas are rare; they include:
 1. **Ganglioneuroma:** Ganglion cells and nerve fibers in connection with sympathetic ganglia.
 2. **Neuroblastoma:** Less differentiated and occurs in infants & young children. Disseminated by the blood stream.

Neurofibroma

It arises from the connective tissue of the nerve sheath and could be localized or generalized.

1. Localized

- It is usually found in the SC tissue.
- Mobile sideways (at a right angle to the nerve from which it arises, but not along the nerve)
- The patient may C/O parasthesia or pain.
- Cystic degeneration or sarcomatous changes may occur.

2. Generalized (*von Recklinghausen's disease*)

- Inherited (autosomal dominant-AD) disease.
- Any cranial, spinal, or peripheral nerve may be diffusely or nodularly thickened.
- Overgrowth occurs in connection with the endoneurium.
- It is associated with pigmentation (*café-au-lait*) of the skin.
- Sarcoma may occur (5%).
- *Plexiform neurofibromatosis* (5th cranial nerve association) & *elephantiasis neuromatosa* (skin thickened, coarse & dry) are generalized forms.



Figure 8. von Recklinghausen's disease (generalized neurofibromatosis)



Figure 9. A Café au lait pigmentation

D. Epidermal Tumors

Tumors arising from the epidermis (epidermal tumors) include the following:

1. Seborrheic keratosis
2. Actinic (solar) keratosis
3. Keratoacanthoma
4. Bowen's disease

E. Moles (Pigmented Nevi)

Synonyms

- "Mole" means "shapeless mass". "Nevus" means a lesion present since birth (birth mark).

Melanocytes

- The melanocyte is generally believed to be derived from the neural crest.
- In normal skin, they appear as clear cells in the basal layer of the epidermis.
- They may ↑ in number in the layers of the skin to form benign pigmented nevi (moles).

Definition

- Moles are benign skin tumors developing from *melanoblasts*, which lie in the basal layer of epidermis (benign abnormality of melanocytic system).

Simple Melanocytic Nevi

- *Types*
 1. **Lentigo**: present within the basal layer of the epidermis.
 2. **Junctional**: as localized aggregations projecting into the dermis.
 3. **Dermal**: entirely within the dermis.
 4. **Compound**: show the features of both the junctional and dermal nevi.
- *Incidence*: One or more melanocytic nevi are present in over 95% of whites.
- *Age*: Any age, but commonly in childhood and adolescence.
- *Site*: Anywhere in the skin, but mostly on limbs and face.
- *Clinical presentation*: It presents as a small, brown, flat or slightly raised (junctional or compound). With increasing age they either atrophy or mature into dermal nevi.

The Blue Nevus

- *Definition*: It is a tumor of *dermal* melanocytes i.e. lies deep in the dermis.
- *Site*: Commonly found on the face and dorsum of hands.
- *Clinical presentation*: Very darkly pigmented lesion, but the thinned overlying epidermis masks the blue color of melanin & gives the nevus its characteristic shiny blue color
- *Complications*: Rarely changes into malignant melanoma.

Congenital Hairy Nevus

- Congenital nevi are rare.
- Often darkly pigmented, may be hairy and papillary in appearance.
- Occasionally they cover extensive areas of the skin.
- May undergo malignant change even in childhood.

Halo Nevus

- This term is applied to a nevus that develops an area of surrounding *leukoderma*.
- It may be followed by gradual disappearance of nevus.
- Its clinical importance lies in its possible confusion by clinicians with malignant melanoma.
- No therapy is required, unless clinical doubt exists. If excision is performed, the hypo-pigmented area should be included.

Treatment of Nevi

- Nevi are virtually always benign before puberty.
- Treatment may be indicated for (1) cosmetic reasons, (2) if the nevi are subject to trauma, or (3) if there are alerting signs of change.

Alerting Signs, Symptoms and Conditions

1. *Giant pigmented nevus or bathing trunk nevus*: treatment is difficult because of the extent and potential threat of melanoma developing. Surgical excision and SSG coverage may help.
2. *Rapid ↑ in size*. It becomes wider and thicker, often changing from a flat plaque to a nodule
3. *Ulceration and bleeding*. As tumor cells multiply, the overlying epithelium becomes anoxic and either ulcerates spontaneously or breaks down after a very minor injury. The subsequent bleeding is mild, but recurs each time the scab is rubbed off.
4. *Itching*.
5. *Fixation*.
6. *Involvement of adjacent structures*.
7. *Dissemination*.
8. *Change in consistency*.
9. *Change & deepening of color* (malignant melanocytes usually produce more melanin).
10. *Halo*: The pigment produced by malignant melanocytes may spread diffusely into the surrounding skin to produce a brown halo around the 1st lesion.
11. *Satellites*: Malignant cells may also spread via the skin in intra-dermal lymphatics. When they stop migrating & multiply they become small intra-dermal nodules around the 1st lesion (satellite nodules).
12. *FAMMM (familial atypical multiple-mole-melanoma syndrome)*: Dysplastic nevi in families with large numbers & ↑ incidence of malignant melanoma. It is an AD disorder with variable incomplete penetrance of the clinical phenotypes. Patients with the FAMMM syndrome are genetically loaded with an ↑ risk of developing melanoma & other malignant neoplasms. It requires careful monitoring with sun protection. Clinically, dysplastic nevi appear as flat macules or thin “pebbly” plaques that display ABCD as shown in Table 3.

Table 3. Recognition of melanoma according to the American ABCD system

American ABCD system	Glasgow 7- Point Checklist	
	Major Features	Minor Features
• <i>Asymmetry</i>	Diameter > 6mm	Inflammation
• <i>Border (irregular)</i>	Change in shape	Oozing or bleeding
• <i>Color (irregular)</i>	Change in color	Mild itch or altered sensation
• <i>Diameter (> 1 cm)</i>		

PRE-MALIGNANT SKIN LESIONS

1. *Actinic (solar) keratosis*. Prolonged exposure to sunlight results in areas of hyper-keratosis of the skin, which may turn malignant.
2. *Chronic radiation dermatitis*: Excessive exposure to irradiation in high doses results in ulceration. Eventually squamous cell carcinoma (SCC) may develop.
3. *Bowen's disease* (carcinoma in-situ between the epidermis & dermis).
4. *Leukoplakia* (Figure 10)

It means "any white patch or plaque that cannot be characterized clinically or pathologically as any other disease". In the oral mucosa, it varies from a small circumscribed lesion to an extensive one. Color may be white, yellowish, or grey. It is soft or thick and crusty, smooth or wrinkled with cracks and fissures. Induration suggests malignant change and is an indication for immediate biopsy. It is important to recognize that the speckled or nodular leukoplakias are the most likely to undergo malignant change. Etiology is mainly smoking.



Figure 10. Leukoplakia of the tongue. Eventually, a SCC may develop.

5. *Keratoacanthoma* (Figure 11)

It results from overgrowth & subsequent necrosis of a sebaceous gland due to unknown causes. Synonyms include "Molluscum sebaceum, adenoma sebaceum, Molluscum pseudo-carcinomatousum". "Molluscum" means "soft". It presents with a conical lump (1-2 cm) with a necrotic brown or black center (looks like a volcano). It usually affects the face and nose. The bulk of the lesion is firm & rubbery, but the central core is hard in consistency



Figure 11. Keratoacanthoma

6. *Xeroderma-pigmentosum* (Figure 12)

It is a rare genetic disease transmitted via an incomplete sex-linked recessive gene and occurs primarily as a skin condition. Patients exhibit extreme sensitivity to sunlight from an early age with defective DNA repair with UV light exposure. The disease begins in early childhood with diffuse freckling, progressive drying & thinning of the skin. By early adult life, malignant skin changes occur. Death occurs from metastatic disease. Treatment is by absolute protection from sun exposure & early aggressive treatment of skin tumors as they occur. Recurrence is common.



Figure 12. Xeroderma-pigmentosum

7. Chronic ulceration and unstable scars

Long standing irritation such as (1) unstable burn scars (*Marjolin's ulcer*), (2) draining sinuses or fistulas, (3) varicose ulcers, (4) vaccination scars, or (5) tattoos. Anaplastic change occurs only after many years. Diagnosis is often delayed. Classically, the lesion is painless & grows slowly, with rare 2^{ty} deposits. Any type of malignancy can occur.

Marjolin's ulcer (Figure 13)

- First described by *Jean Nicholas Marjolin* in 1828.
- Marjolin's ulceration is a SCC arising at sites of chronic inflammation.
- Recognized underlying causes include chronic venous ulceration, burns and osteomyelitis sinuses.
- Usually a long-period of time (10-25 years) exists between injury and malignant transformation. Approximately, 40% occur on lower limb.
- Malignant change is usually painless.
- Nodal involvement is uncommon.
- Diagnosis is confirmed by biopsy of the ulcer edge.
- Treatment: Adequate excision and skin-grafting, or amputation



Figure 13. Marjolin ulcer (SCC) on top of a chronic burn scar.

8. Junctional Nevi (→ Malignant Melanoma)

MALIGNANT SKIN LESIONS (SKIN CANCER)

Etiology of Skin Cancer (Predisposing Factors)

1. *Actinic (UV) rays*: They cause DNA damage and T-helper cell suppression. It is the most important factor. It is not surprising that 80-90% of malignant skin tumors occur in the head and neck, and dorsum of hands (exposed areas).
2. *Occupation (outdoor occupations)*: Sailors, farmers, engineers (exposed to sunrays).
3. *Ethnic factor (race)*: Fair-skinned Caucasians with light hair and eye color are at greater risk.
4. *Genetic (inherited) factor*: such as Xeroderma and von Recklinghausen's disease
5. *Physical factor*: Ionizing radiation, due to ionization of important cell constituents.
6. *Chemical exposure*: Tobacco-associated nitrosamines and chronic exposure to arsenic, coal and tar
7. *Mechanical factor*: Chronic irritation by dentures, or osteoma, etc.
8. *Immunosuppression*: Incidence is more in renal transplant recipients.

Cancer is a disease of genes

There are three important classes of genes involved in cancer:

1. *Tumor suppressor genes*: Control the cell cycle by slowing it down.
2. *Oncogenes*: Promote cell proliferation.
3. *Growth factors & their receptors*: Switched-on by oncogenes or switched-off by tumor suppressor genes.

BASAL CELL CARCINOMA (BCC)

Definition

- It is a “*locally invasive*” carcinoma arising from pluripotential cells within the *basal layer* of the epidermis or follicular structures. It *rarely* metastasizes (<0.1%), but *may kill* by local infiltration

Etiology of BCC

1. Ultraviolet (UV) radiation.
2. X-irradiation.
3. Arsenic exposure
4. Immunosuppression.
5. Xeroderma-pigmentosum
6. Gorlin’s syndrome: multiple BCCs occur in this AD condition, often at an early age.

Incidence

- *General incidence:* It is the most common malignancy in humans and the commonest malignant tumor (MT) of the skin (60%).
- *Age:* It usually occurs in adulthood especially elderly people (mostly > 65 years - related to duration of exposure to UV light).
- *Gender:* Men > women (2:1) because of working more out of doors.
- *Geography:* It is more common in countries that have much bright sunlight.
- *Ethnic Group:* It affects mainly Caucasians & fair-skinned people (e.g. Australia), and is rare in dark-skinned races.

Symptoms

- A persistent *nodule or ulcer (often multiple)*, with a central scab that repeatedly fall off and then reforms, giving the patient a false impression that it is benign and not important.
- *Disfigurement.*
- It may cause *itching*. If neglected and becomes deep, it may cause *pain, bleeding*, and may become *infected*.
- The large neglected rodent ulcer *destroying* one side of the face is nowadays, fortunately, rare

Clinical Examination

- **Site:** It typically occurs in areas of chronic sun exposure; so >85% occur in head & neck (face, ears, scalp, neck), or arms and hands i.e. It is (a) *facial* including the scalp or (b) *extra-facial* in the neck, arms & hands (exposed areas). The back of the ear, upper eyelid and lower lip are rarely affected.
- Slowly growing.
- Can cause significant local destruction and disfigurement if neglected.
- It is often very friable and prone to ulcerate, providing a nidus for infection.
- Local L.Ns should *not* be enlarged (unless infected or transformed into a SCC).

Clinical Subtypes of BCC

1. Nodulo-ulcerative BCC (Figure 14)

- It is the most common variety of BCC.
- It starts as a firm waxy *papule* or as a white *nodule* with a pearly appearance.
- Telangiectatic vessels are seen over the surface.
- Later on, the center becomes atrophic and then ulcerated, leaving the typical, *raised, rolled edge*.
- There may be bleeding and crusting
- Translucency
- It may take as long as one year to double its size. It spreads slowly with erosion of deeper structures(*rodent ulcer*)



Figure 14. Nodulo-ulcerative BCC. Note the raised, rolled-in edge of the ulcer

2. Pigmented BCC

- It has the same previous features of nodular BCC, but with ↑ brown or black pigmentation. That is why it may be confused with malignant melanoma.
- It is common in individuals with dark skin.

3. Cystic Type

- Translucent blue-grey cystic nodules.

4. Superficial Type

- It is characterized by scaly patches or papules; pink to red brown in color.
- Erosion is less common as it has a little tendency to become invasive.

5. Morphea-like (Fibrosing BCC)

- It presents as a slightly elevated, firm, yellowish plaque.
- The skin over remains intact for a long time
- The border is ill-defined because tumor cells invade normal tissue well beyond the visible margin. It, therefore, requires a wider safety margin on excision than the typical BCC.

6. Cicatricial

- The skin actually looks shiny and taut because of the intense fibroblastic response the tumor induces giving its scar-like appearance.

1. Basi-squamous Type

- It has the general configuration of a BCC, but contains also atypical squamous cells.
- It is more aggressive than the typical BCC.

Diagnostic Investigations

- Incisional biopsy (in large lesions). It is taken from the edge and should include part of the lesion and normal skin around it.
- Excisional biopsy: The tumor cells tend to align more densely in a *palisade* pattern.
- CT scan to evaluate depth of invasion.

Differential Diagnosis of BCC

1. Actinic keratosis.
2. Bowen's disease.
3. Fibrous papules.
4. Keratoacanthoma (It has a shorter history and deep slough than BCC)
5. Melanocytic nevi (if BCC is pigmented)
6. Seborrheic keratosis
7. Squamous cell carcinoma (SCC) (It has a shorter history than BCC and a characteristically everted edge).

Complications

1. *Spread* by direct infiltration of muscles, cartilage, and bone (locally malignant).
2. *Secondary infection* (local lymph nodes become enlarged & tender).
3. *Hemorrhage*: From erosion of a blood vessel. It may be severe.
4. *Epitheliomatous transformation* (into a SCC) which is evidenced by:
 - a) Growth becomes rapid.
 - b) Everted edges at least in a part of the ulcer.
 - c) Induration extends beyond the base.
 - d) Loss of the pearly white margin of the BCC.
 - e) An enlarged LN, which becomes hard & fixed.
 - f) Evidence of distant metastases.
 - g) Biopsy (the surest diagnosis).

Treatment

1. *Surgical excision* with a safety margin followed by reconstruction with skin grafts or flaps is very effective. Cure rates reach 85-95%. Margins of 2-5 mm are adequate. In larger tumors with longer history, margin should be 1 cm. In recurrent BCC even larger margins are required.
2. *Radiotherapy*: BCC is very radiosensitive. The response rate is 92 %. It is reserved for the elderly who are not suitable for surgery or in specialized anatomical sites.
3. *Cryotherapy*: Using CO₂, or liquid nitrogen (for very small lesions).
4. *Moh's micrographic surgery (chemosurgery)*: This involves injection of a chemotherapeutic agent then serial horizontal excisions and mapping of the tumor so if any foci of tumor is detected by frozen section, further excision is carried out.
5. *Topical 5-Fluorouracil (5-FU)*: It has *no place* in invasive types. It is only useful in patients with extensive sun damaged skin.

Prognosis

- Prognosis is excellent with proper therapy.
- About 20% will have a 2nd lesion within 1 year (follow-up should be every 6 month-1 year).

SQUAMOUS CELL CARCINOMA (SCC)

Definition

- Squamous cell carcinoma (SCC) (epithelioma, epidermoid carcinoma) is a carcinoma of the *prickle cell layer* of the epidermis that normally migrates outwards to the surface to form the superficial keratinous squamous layer.
- It arises from *epidermal keratinocytes*. The tumor cells infiltrate the epidermis, dermis and adjacent tissue.

Incidence

- *General incidence*: It is the 2nd common cancer of the skin.
- *Age*: The incidence of SCC ↑ with age (usually >50 years).
- *Occupation*: Incidence ↑ with prolonged exposure to sunlight (sailors) and certain chemicals (engineers).

Etiology

- Arises in an area that has had some *pre-malignant change* (refer back).
- UV sunlight exposure - Therapeutic UV exposure (strong correlation with damage of the skin by the sun) - DNA repair failure
- Ionizing radiation
- Fair complexion
- Chemical carcinogens (e.g. arsenic, coal, tar).
- Immunosuppression (iatrogenic or non-iatrogenic)
- Chronic inflammatory and scarring conditions such as syphilis (\$), lupus vulgaris, leprosy, chronic ulcers, osteomyelitis, hydra-adenitis suppurativa, lymphoedema, long-standing venous ulcers and old burn scars
- Some cases are *de-novo*.

Pathology

- *Macroscopic picture*: A typical ulcer with everted edges and necrotic floor.
- *Microscopic picture*: Tongues of malignant cells in all directions with clusters (+ve for keratin). Cell nests in cut sections in the ramifications of the tumor appearing as rounded masses formed of cuboidal cells (peripheral) with no palisading (characteristic of BCC), prickle cells (middle) and keratin (central). Malignant cells show pleomorphism and loss of polarity. The nucleus shows hyperchromatism, increased mitotic figures and multiple nucleoli.
- *Spread*: Local infiltration, lymphatic and hematogenous (very rare and late, mostly to the lungs).

Symptoms

1. The patient C/O bleeding & discharge from an ulcer (Figure 15), or of a lump (fungating fleshy or pinkish nodular lesion). It may be multiple. It may occur on chronic scars due to burns (Marjolin ulcer)
2. It may become painful if it invades deep structures or becomes infected.
3. The patient may complain of enlarged LNs or systemic symptoms and be unaware of the 1st lesion.

Physical Examination (Figure 15)

- **Site:** It can occur on any part of the skin but is more common on exposed skin and skin subjected to repeated chemical or mechanical irritation. The head and neck region is the most common site (70%) especially the lower lip, ear and cheeks. About 15% is found on the upper extremities.
- **Color:** The everted edge of the ulcer is usually dark red-brown in color because it is very vascular.
- **Shape and Size:** SCC begins as a small nodule. As it enlarges the center becomes necrotic, sloughs and the nodule turns into an ulcer, which is initially circular but can become of any shape as it enlarges.
- **Discharge:** If it becomes infected, the discharge becomes copious, bloody, purulent and foul.
- **Floor:** The floor of the ulcer may be covered with old coagulated blood, serum, or *necrotic material*. Surface changes may include scaling, ulceration, crusting, or the presence of a cutaneous horn.
- **Edge:** *Everted* because excessive tissue growth raises it above and over the normal skin surface.
- **Base:** It is *hard and indurated*.
- **Depth:** Soft tissues are easily invaded and when they slough, they leave a deep ulcer.
- **Relations:** Depend on the extent of malignant infiltration which causes the ulcer to become *fixed*. A background of severely sun-damaged skin, including solar elastosis, mottled dys-pigmentation, telangiectasia and multiple flakes is often noted.
- **Local LNs:** These are often *enlarged* (from metastases). Later, they become hard and fixed. Regional LNs may become enlarged also as a result of infection of the ulcer.
- **Complications:** Infection and bleeding re the most common. Bleeding may be massive.
- **General examination:** For detection of distant metastases (rare) e.g. lung collapse, or pleural effusion caused by pulmonary metastases



Figure 15. SCC with the characteristic everted edge & necrotic, bleeding floor.

Investigations

- **Biopsy** for diagnosis and grading.
- **Plain X-ray** of the related bone.
- **Investigations for distant metastases** a suspected (lung, bone, liver, etc).

Differential Diagnosis

- Actinic keratosis
- *Basal cell carcinoma.*
- Keratoacanthoma
- Infected seborrheic warts
- Pyoderma gangrenosum (*pyogenic granuloma*)

SCC versus BCC

- SCC shares with BCC similar causes, anatomic sites, and age incidence SCC.
- It is less common than BCC with a ratio of (1:4)
- It is more aggressive than BCC.
- It grows more rapidly
- It tends to ulcerate earlier
- It does metastasize in a small percentage
- There are 2 types: a slowly growing exophytic type and a rapidly growing ulcerative type.
- The ulcer is irregular, with raised everted edges, necrotic floor and an indurated base with later fixation.
- Regional LNs are often enlarged.

Treatment

1. *Surgical excision*
 - It is the most appropriate treatment.
 - *The safety margin* for a curative excision is judged by the visible and palpable extent of the tumor and should be approximately 1 cm, ↑ as the size of the tumor ↑. In more extensive lesions a 2-3 cm (one inch) margin may be necessary.
 - *Regional LN dissection* is required if proved to be metastatic.
2. *Radiotherapy*
 - It is used for massive, unresectable tumors in critical anatomical sites. Postoperatively, it may be used for persisting tumor or where clearance is doubtful.
 - It is also used to treat enlarged *fixed* LNs.
3. *Systemic chemotherapy*: For metastatic lesions.
4. *Cryotherapy*: For selected lesions.
5. *Moh's micrographic surgery*: For selected lesions.

Prognosis

- *Size of the tumor*: If small (< 0.8 mm in diameter) → a cure rate of 95%.
- *Lymph Nodes*: If involved → poor prognosis.

MELANOMA

Definition

- Melanoma is a malignant neoplasm arising from melanocytes (mature melanin-forming cells)

Embryology of Melanocytes

- Melanocytes are derived from the *neural crest tissue*.
- Cells migrate during early gestation to the *skin, uveal tract, meninges and ectodermal mucosa (oral cavity, esophagus, vagina and anal canal)*.
- In the skin, melanocytes reside in the *basement layer of the epidermis* and elaborate *melanin* under a variety of stimuli.

Incidence

- *General incidence*: It is the 3rd common malignant tumor of the skin. It is *no* longer "rare".
- *Age*: Rare before puberty but can occur in children. It is most frequent between 30-50 y.
- *Gender*: No sex predilection; both sexes are affected equally.
- *Ethnic group*: It is common in Caucasians and rare in Negroes.
- *Geographical distribution*: It is more common in sunny areas such as Australia and New Zealand, particularly in fair-skinned people with light complexion.
- *Occupation*: Those who work outdoors, in excessive sunlight, are particularly susceptible
- *Multiple melanomas*: Melanoma is usually *solitary*, often with multiple 2^{ry} nodules around the 1^{ry} lesion. Multiple melanomas are very rare (1-4% of all melanomas).
- *Familial melanoma*: In familial cases, the incidence of multiple lesions may reach 20%.
- *Extra-cutaneous melanoma*: Melanomas occurring in the eye, meninges or mucocutaneous junctions such as the anus & mouth are far less common than cutaneous melanoma (rare).

Etiology (Risk Factors of Melanoma)

- *Family* or personal history (e.g. FAMMM syndrome)
- *Fair* (blond or red) hair - Blue, green, or grey eyes
- *Freckling* of upper back
- *Sunlight*: Presence of actinic keratosis - Three or more blistering sunburns before age 20 years.
- *Premalignant lesions*: *Nevi* (dysplastic congenital nevi, large numbers of common nevi (>100), large congenital nevi (>20 cm), changing mole) and xeroderma pigmentosa.
- Albinism
- Immunosuppression
- Radiotherapy (RT)
- Previous malignancy

Pathology

Clark's Level

The atypical proliferation of intra-epidermal melanocytes (radial growth phase) precedes the dermal invasion (vertical growth phase).

- Level I: in-situ confined to epidermis.
- Level II: invasion into papillary dermis.
- Level III: To junction of papillary and reticular dermis.
- Level IV: invasion into reticular dermis.
- Level V: invasion into subdermal fat.

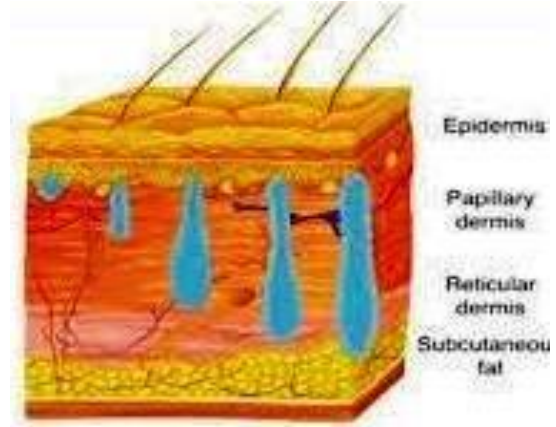


Figure 16. Melanoma (Clark's levels)

Breslow's Thickness

- Depth of invasion as measured by thickness is the most important parameter.
- Done using an optical micrometer.
- Lesions < 0.76 mm in thickness have a very favorable prognosis

Spread

1. *Local spread*: To surrounding structures.
2. *Lymphatic spread*: By embolization to form LN secondaries or permeation to form satellites, or in-transit metastases. The latter result from melanoma cells being trapped between the 1st tumor and regional LNs. This produces a region of cutaneous metastases located >3 cm from the 1st site.
3. *Blood spread*: To the lungs, liver and brain, or other tissues. It may lead to *melanuria*.

Symptoms

1. *Changes in a previous mole*: Change in color (erythema around it), change in size (rapid rate of growth), ulceration, bleeding, satellite nodules, itching, local LN or distant metastases.
2. *Cosmetic disfigurement*. Melanoma is usually in the form of a *nodule or ulcer*.
3. *Itching* (but not pain).
4. *Evidence of distant metastases*, such as weight loss, dyspnea, or jaundice.

Physical Examination

- *Site*: The majority is found in limbs and head and neck (Figure 17). The commonest site in women is the lower leg and in men the trunk (front and back). It may occur at mucocutaneous junctions (mouth and anus).
- *Color*: It varies from pale pinkish-brown to black. If they have a rich blood supply they develop a purple hue.
- *Shape and size*: It can form a florid tumor, protruding from and overlapping the surrounding skin



Figure 17. Melanoma of the forehead

- *Surface:* When small, it is covered by smooth epithelium. When the epithelium dies from ischemic necrosis, the resulting ulcer is covered with a crust of blood and serum. Bleeding and subacute infection may make the surface of the tumor wet, soft & boggy.
- *Consistency:* The 1^{ry} tumor is firm, but satellite nodules feel hard.
- *Relations:* The malignant tissue is intimately fixed to the skin.
- *Regional LNs:* When involved, they are enlarged, hard, painless, mobile or fixed.
- *Surrounding tissues:* There may be a *halo*, or *satellite nodules* around the 1^{ry} lesion. If the tumor has been itchy, the surrounding skin may be excoriated.
- Differences in the local features between malignant (melanoma) and benign pigmented lesions are demonstrated in Figure 18.
- *General examination*
- Melanomas may spread to the lungs, liver and brain, resulting in *pleural effusions*, *hepatomegaly*, *jaundice* and *neurological abnormalities*, respectively.

Clinical Types) of Melanoma

1. *Superficial spreading melanoma:* It is the most *common* (60-70%), intermediate in malignancy, usually palpable, with variegated color & irregular edge. It affects mainly the legs in women & the back in men.
2. *Nodular Melanoma:* It is the most *malignant*, represents 12-25% of cases, occurs in younger age groups, usually convex in shape, well-delineated, with a uniform color (blue, grey or black), smooth surface and early ulceration.
3. *Lentigo maligna melanoma (Hutchinson's melanotic freckle):* It is the least *common* & least malignant. It represents 7-15% of cases. It occurs mostly on the face, head and neck of old people (>60 years), and in women > men in patients. It begins as irregularly pigmented flat brown macule that grows very slowly (1-15 years).
4. *Acral lentigo melanoma:* It has a poor prognosis and occurs on the palms, soles and in sub-ungual locations.
5. *Amelanotic melanoma:* It may be pink and often present with LN metastases. It has a very bad prognosis.

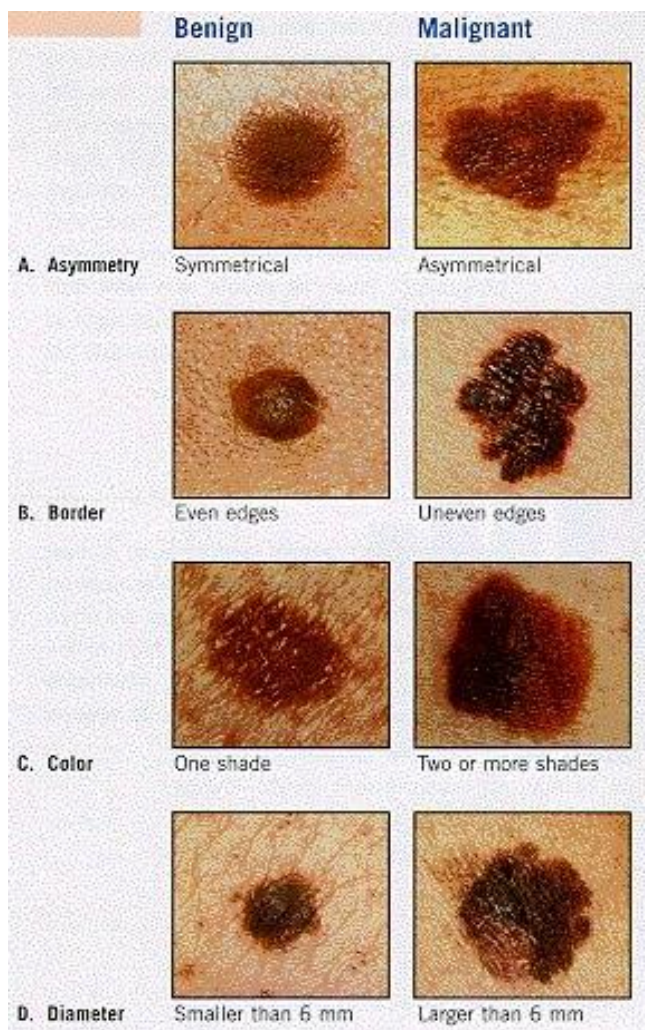


Figure 18. Local differences between benign pigmented lesions and melanoma (malignant)

Staging of Melanoma

- Clark's level
- Breslow thickness
- AJCC staging: Table 4 summarizes the staging of melanoma

Table 4: AJCC staging of melanoma

Stage	Breslow	Clark
Ia	<0.75mm	Clark's II
Ib	0.76-1.5mm	Clark's III
IIA	1.51-4mm	Clark's IV
IIB	> 4mm	Clark's V
III	regional LN	in-transits
IV	systemic metastases	systemic metastases

Investigations

1. Biopsy
 - Suspicious lesions (moles) should be excised completely with a 2-mm margin.
 - No incisional or punch biopsies should be attempted.
 - Excisional biopsy is necessary for diagnosis and micro-staging (degree of depth and thickness), which is necessary for planning treatment. Biopsies are taken from suspected moles + sentinel LN
2. Labeled-monoclonal antibodies (for detection of deposits).
3. Serum LDH levels + Serum S-100 protein levels.
4. CT-scan & investigations for detection of suspected distant metastases.

Differential Diagnosis

Malignant melanoma has to be differentiated from **other pigmented lesions**, mainly the following:

- *Moles (pigmented nevi)*
- *Pigmented BCC*: Most common in middle age, blue black in color, raised edges and capillary neovascularity.
- *Seborrheic keratosis*: Occasionally black in color, usually 1cm in diameter or larger, typically appear raised, warty, greasy and as if being “stuck onto the skin”.
- *Dermatofibroma*: Occasionally dark brown in color, usually smooth, slightly raised and never contains hair. It grows very slowly and never becomes malignant
- *Pyogenic granuloma*: A rapidly growing, pink, nodule that results from minor trauma
- *Nevus of Ota (congenital melanosis bulbi)*: It is a blue hyper-pigmentation caused by entrapment of melanocytes in the upper 1/3 of the dermis. It is found on the face unilaterally and involves the first 2 branches of the trigeminal nerve. The sclera is involved in 2/3 of cases (causing ↑ risk of glaucoma). Women are affected > men (5:1), and it is rare among Caucasian people. Nevus of Ota may not be congenital and may appear during puberty.
- *Becker's nevus (Becker's melanosis or Becker's pigmentary hamartoma)*: It is a skin disorder predominantly affecting males. The nevus generally first appears as an irregular pigmentation (melanosis or hyper-pigmentation) on the torso or upper arm and gradually enlarges irregularly, becoming thickened and often hairy (hyper-trichosis). The nevus is due to an overgrowth of the epidermis, pigment cells (melanocytes), and hair follicles.
- *Epithelial polyp (Pinkus tumor)*: It is a histological variant of superficial BCC and shows a profound stromal component. Clinically, they are mostly located in the elderly on the trunk & lower extremities. The tumor grows, minimally invasive and is rare.
- *Dermatosis Papulosa Nigra (DPN)*: It is a condition of many small, benign skin lesions on the face, a condition generally presenting on dark-skinned individuals. It is extremely common, affecting up to 30% of Black people in the USA. Histologically, DPN resembles seborrheic keratosis. The condition may be cosmetically undesirable to some patients.
- *Bowen's disease* and Bowenoid papulosis.

- *Sub-ungual hemorrhage*: It is usually sudden in onset and sharply defined under the nail bed. In contrast, melanoma is of gradual onset and defined by poorly demarcated streaks extending along the axis of the nail and evacuating the blood. With the passage of time, the entire subungual hemorrhage will migrate distally with clearing of the nail bed. Subungual melanoma, however, is a persistent lesion
- *Skin pigmentation with other diseases*: Cafe au lait patch and multiple circumoral moles associated with Peutz-Jegher syndrome.

Treatment of Melanoma

1. Treatment of the 1st lesion (surgical resection)

- A safety margin of 1-2 cm for thin (<1mm) lesions and 3 cm for thicker ones is adequate.
- No excision beyond deep fascia is done.

2. Treatment of LNs

- FNAC of LNs should be performed to confirm diagnosis.
- Lymphatic mapping and sentinel LN biopsy (SLNB) (intra-dermal injection of radioactive colloid)
- Therapeutic LN dissection for clinically positive nodes requires radical clearance.

Sentinel LN biopsy (SLNB)/dissection

- Lymphatic mapping and SLNB have effectively solved the dilemma of whether to perform regional lymphadenectomy (in the absence of clinically palpable nodes) in patients with thicker melanomas (≥ 1 mm in depth).
- SLNB for cutaneous melanoma was developed in the early 1990s to allow a selective approach to identifying individuals with occult regional nodal metastasis through localization of the first-draining, or sentinel LN (Morton, 1992).
- The success of the technique is based on the concept that cutaneous lymphatic flow is well-delineated in melanoma and that the histology of the sentinel node is characteristic of the entire LN basin (i.e. a -ve sentinel LN obviates the need for further LN dissection) (Morton, 1999).
- Pre-operative radiographic mapping (lympho-scintigraphy) and vital blue dye injection around the 1st melanoma or biopsy scar (at the time of wide local excision/re-excision) is performed to identify and remove the initial draining regional LN(s).
- The sentinel LN is examined for micro-metastasis using routine histology and immuno-histochemistry; if present, a therapeutic or completion LN dissection (CLND) is done.
- A -ve SLNB result prevents the morbidity of an unnecessary lymphadenectomy.

3. Treatment of disseminated disease – loco-regional recurrence

- *Surgery*: Palliative for:
 - a) Removal of painful or ulcerated SC metastases
 - b) Relief of intestinal obstruction.
 - c) Resection of isolated pulmonary or cerebral metastases
 - d) Debulking prior to radiotherapy or chemotherapy.

- *Radiotherapy (RT)*: Malignant melanoma is a highly *radioresistant* tumor. However, RT has a place in the treatment of:
 - a) Fixed inoperable LNs
 - b) Cerebral and bone metastases
 - c) Cutaneous recurrence not responsive to other forms of treatment.
- *Systemic chemotherapy*
- *Immunotherapy*

Prognosis of Melanoma

Negative prognostic criteria of melanoma include the following:

1. Thickness > 0.76mm
2. Clark's level > II
3. Older age
4. Male gender
5. Satellites
6. Ulceration
7. Regional LN involvement
8. Distant metastases

SKIN SARCOMAS

- Skin sarcomas can develop from fatty tissue, vascular tissue and fibrous tissue present in the dermis.
- Common examples are:
 1. Liposarcoma
 2. Neurofibrosarcoma
 3. Dermatofibrosarcoma protuberans
 4. Kaposi's sarcoma.
- Treatment usually entails surgical resection followed by chemotherapy and radiotherapy.
- Radical resection may deem amputations necessary in many situations.

MYCOSIS FUNGOIDES (LYMPHOMA)

- Mycosis fungoides is an uncommon chronic T-cell lymphoma primarily affecting the skin.
- It may appear as a rash, nodules, or ulceration. The lesion may be multiple.
- It is treated by electron beam radiotherapy (EBRT).

SKIN ULCERS

Etiological Classification

1. Non-Specific <i>Traumatic / burns</i> - Mechanical - Physical - Chemical <i>Vascular</i> - Arterial - Venous - Lymphatic <i>Neurogenic</i> <i>Metabolic (systemic)</i>	- Pressure of splint or dental ulcer of the tongue. - Electric or X-ray burn. - Caustics. - Atherosclerosis, Buerger's or Raynaud's disease - Varicose ulcer, post-phlebitic ulcer. - Lymphedema - Decubitus (pressure sores) - Neurotrophic: e.g. cord lesions & peripheral neuropathies. - Gout (ulceration of a tophi) - Diabetes mellitus
2. Specific	- TB, Syphilis (\$) - Soft sore - Actinomycosis - Parasitic (Leishmaniasis)
3. Malignant	- 1^{ry} skin tumors (SCC, BBC, melanoma, sarcomatous ulcer) - Metastatic ulcer

General Management of Skin Ulcers

1. Diagnosis of cause and its treatment
2. Rest and elevation of the extremity affected
3. Local care of the wound
4. Reconstruction whenever possible
5. Recent topical applications:
 - Calcium alginate (Kaltostat) - Carboxymethylcellulose (Aquacell)
 - Ozone therapy
 - Low intensity direct current
 - Topical opioids
 - Micromechanical forces
 - Topical human epidermal growth factor (GF) Fibroblast GF spray
 - Electrolyzed super-oxidized solution

CHAPTER

8

Aesthetic Surgery

Despite the popular misconception, the word "plastic" in "plastic surgery" does not mean "artificial," but is derived from the ancient *Greek word "plastikos"*, which means "to mold or give form". Plastic surgery includes both the reconstructive & aesthetic subspecialties.

RHYTIDECTOMY (FACE-LIFT)

Introduction

- In the younger individual, the face is firm and smooth due to fatty tissue directly beneath the skin, which fills out the contours of the face and gives it an even, rounded appearance.
- As people grow older, the natural aging process, genetic influences, exposure to the sun and the effect of gravity cause the skin to wrinkle and sag, particularly around the chin, on the jaw line and on the neck.
- Pathway to photo-aging: The slight imperfections caused by UV damage accumulate over time and the micro-scars become the macro-scars, which we call wrinkles.

Indications of Face Lift

1. Forehead and glabellar creases
2. Ptosis of the lateral eyebrow
3. Deepening of the naso-jugal groove and palpebral-malar groove
4. Ptosis of the malar tissues
5. Generalized skin laxity
6. Downturn of the oral commissures
7. Deepening of the labio-mental creases
8. Jowls (Drooping of the lower part of cheeks / loose fleshy part of the neck)
9. Loss of neck definition and excess fat in neck
10. Platysmal bands

Operative Procedure (Standard Face-lift)

- *Incisions* are made inside the hairline at the temple, running in front of the ear then around the earlobe and behind the ear, ending in the hair of the scalp (Figure 1).
- *Undermining*: Loose skin is separated from underlying tissue and is pulled upwards and backwards, and excess skin is excised (Figure 2).

- **Redraping:** Connective tissue and sagging muscles are tightened (Figure 3). Fat deposits are removed from beneath the chin and neck. This may necessitate an additional small incision under the chin. Tiny sutures are used to close the incisions.
- **Operative time:** A rhytidectomy may take from 3-5 hours depending on whether other procedures are done at the same time.
- **Post-operative care:** Hospitalization usually unnecessary. Variable swelling and bruising are expected and usually resolve spontaneously. *Instructions* include head elevation for several days and control of blood pressure (BP).



Figure 1. Incision within the hairlines



Figure 2. Undermining (upwards and backwards)



Figure 3. Redraping (CT and sagging muscles are tightened)

Complications

- Hematoma formation
- Skin slough
- Hypertrophic scarring
- Nerve injury (posterior auricular, facial).

BLEPHAROPLASTY (EYELID SURGERY)

Introduction

- Eyes themselves are virtually expressionless structures.
- It is the contour of the skin - the tissue, muscle, fat, hair and lashes around the eyes - that conveys expression.
- Wrinkles, deep lines and puffiness of the lids begin to develop with the passage of time.
- *Blepharoplasty* can rejuvenate puffy, sagging or tired-looking eyes by removing excess fat, skin and muscle from the upper and lower eyelids.
- It may be performed for cosmetic reasons or to improve sight by lifting droopy eyelids out of the patient's field of vision.

Operative Procedure

- It can be performed under general anesthesia or local IV sedation.
- It is done through very fine incisions from the inner to the outer edge of the eyelid. In selected cases, incisions can be made inside the eyelid.
- The upper eyelid is worked on first with the incision made in the fold of the lid (Figure 4).
- On the lower eyelid, the incision is made directly below the eyelash (Figure 5). Excess fat and skin are then removed from the underlying compartments.

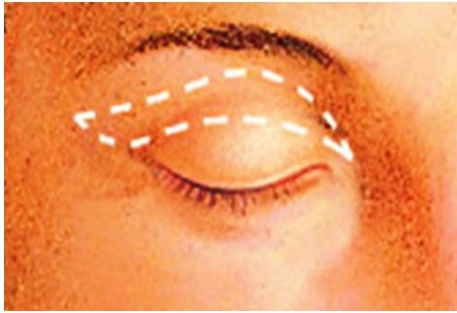


Figure 4. Incision in the fold of the upper eyelid (white lines)



Figure 5. Incision in lower eyelid directly below the eyelash (white line)

- The amount of fat excised is determined by the degree of protrusion of fat when pressure is gently applied to the area.
- Fine sutures are used to close the incision and special ointments and dressings are applied.
- The procedure (Figure 6) takes from 1-2 hours, depending upon the extent of the surgery.



Figure 6. Blepharoplasty. Pre- and post-operative views.

Complications

- Retro-orbital hemorrhage
- Diplopia
- Corneal irritation
- Lower lid ectropion

RHINOPLASTY

Introduction

- The nose is one of the most prominent features of a person's face. Men & women who are dissatisfied with the shape or size of their nose, whether due to natural causes or external trauma to the face, can improve their appearance through a procedure called "rhinoplasty".
- Rhinoplasty is one of the first cosmetic procedures ever developed & is among those most frequently performed today. With rhinoplasty, deformities of the nose are corrected by removing, rearranging or reshaping bone or cartilage; thus, correcting both profile and frontal face views. It may also relieve some breathing problems.

Procedures

- Reshaping the nose by reducing or increasing size.
- Removing hump, changing shape of nose tip or bridge (Figure 7)
- Narrowing the span of the nostrils.
- Changing the angle between the nose and upper lip.



Figure 7.
Rhinoplasty: A young lady with a humpy nasal bridge (left side) corrected with rhinoplasty (right side)

- *Optimum age:* Rhinoplasty is not usually performed until a person has reached the mid-teenage years when growth is nearly complete.
- *Anesthesia:* Local anesthesia with sedation or general anesthesia.
- *Operative time:* 1-2 hours or more – usually as an *outpatient* procedure.

Complications

- Hemorrhage
- Edema
- Excessive narrowing
- Redundant soft tissue
- Nasal bone asymmetry

LIPOSUCTION

Introduction

- Localized stubborn, unsightly bulges can be dealt with "liposuction".
- *In women*, those fat deposits occur most frequently from the waist down, on hips, buttocks and outer thighs (saddle bags).
- *In men*, fat deposits tend to accumulate above the waist, on the abdomen and sides of the waist (love handles).

Indications

- It is designed for those who have specific areas of *localized fat deposits* and who have tried unsuccessfully to eliminate them through diet, exercise and weight loss.
- While the procedure is not designed to correct general obesity, any area where excess fat deposits have accumulated can be treated. These include the chin, neck, jowls, cheeks, arms, inner and outer thighs, buttocks, knees, hips and abdomen.
- Men with enlarged breasts (gynecomastia) can also benefit from this technique.

The Procedure

- *Liposuction* surgery involves using tiny tubes called cannulae (Figure 8) for the removal of fat cells from different sites in the body (Figure 9)

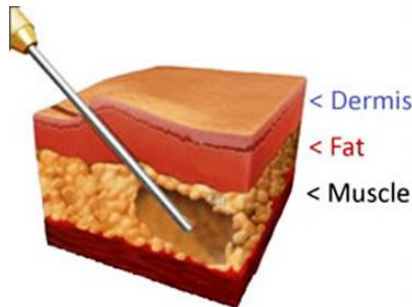


Figure 8. Cannulae of different size and shape used for liposuction



Figure 9. Liposuction can be done for different sites in the body as indicated

- *Gender:* Men & women in good physical condition with good skin elasticity.
- *Age:* When the procedure was first introduced, only younger people were considered to be good candidates; however, recent improvements in the technique have made it possible to treat patients of all ages (Figures 10 and 11).
- *Anesthesia:* It can be done under local anesthesia with IV sedation. When performing what is known as the "tumescent" procedure, this procedure has the advantage of reducing pain and bruising in the area in addition to elimination of blood transfusion.



Figure 10. Liposuction of the infra-umbilical (supra-pubic) region in a young lady



Figure 11. Liposuction of both flanks (and abdomen) in a young gentleman

Post-Operative Care

- A compression garment should be worn after surgery.
- Because liposuction surgery does not involve large incisions and extensive cutting, pain is minimal to moderate and is controlled with oral medication.
- Antibiotics are prescribed to prevent infection.
- Most patients are completely ambulatory following surgery, but rest is recommended for the initial post-operative period.
- Patients are usually able to return to normal activities within a week.

Complications

- Complications connected with this surgery are rare; however, an *uneven skin surface, bleeding, infection, numbness & discoloration* can occur.
- *Contour irregularities* generally fall into three categories:
 1. Over-correction
 2. Under-correction
 3. Failure of skin retraction or abnormal skin retraction

ABDOMINOPLASTY

Introduction

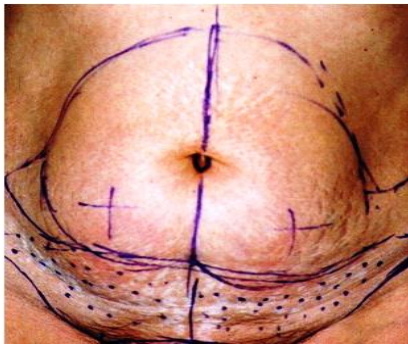
- A *protruding abdomen* as a result of *weak abdominal muscles, weight gain or pregnancy* is a condition that causes distress to thousands of people, particularly that it does *not* respond well to diet or exercise because the skin overlying the muscles has been stretched.
- Abdominoplasty is *not* a substitute for weight loss.
- *The objective of the surgery* is to improve the contour of the body by flattening the protruding abdomen through tightening of abdominal wall muscles and removal of excess fatty tissue and excess skin.
- Thus, the *best candidate for the surgery* is the individual who is of normal weight but who has weak abdominal muscles with excess skin and fat.

Types of Abdominoplasty

1. *Traditional abdominoplasty*: in this procedure, the abdominal wall is exposed, the abdominal muscles are tightened and the skin is pulled down tight with the excess skin being detached. With a traditional abdominoplasty (*tummy tuck*), the entire abdominal area is operated on & the patient's umbilicus (*belly button* or *navel*) is removed (reconstructed by *umbilicoplasty*).
2. *Mini-abdominoplasty*: the patient requires only a small amount of excess skin to be removed, usually due to aging. Compared to the traditional tummy tuck, the mini-tummy tuck is much less invasive. A smaller incision is made and the treatment area consists of the area from the incision to the navel. During the procedure, the navel is *not* moved.
3. *Extended abdominoplasty*: It is an expanded version of the traditional tummy tuck and involves removing excess skin from the patient's flanks, lower back and hips.
4. *Suction-assisted abdominoplasty*: Liposuction of the whole anterior abdominal wall, flap undermining and tightening of the abdominal muscles. The patient's umbilicus is moved (Figure 12).

Complications (Traditional Abdominoplasty)

1. Seroma formation: It is the most frequent complication. It results from wide undermining, which necessitates prolonged post-operative suction drainage to avoid this complication.
2. Increased risk of deep vein thrombosis (DVT) and pulmonary embolism (PE).
3. Skin necrosis: It results from wide undermining of the flaps jeopardizing the blood supply.
4. Sensory changes of the abdomen and thigh.
5. Scar and contour asymmetry: Incisions are often long and geometrically designed and tension may cause excessive scarring.
6. Wound-healing problems/dehiscence.
7. Infection
8. Bleeding/hematoma formation.



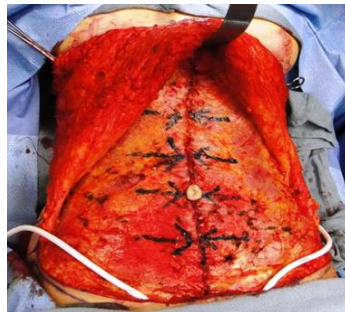
A. A lady with a protruding abdomen, weak AAW and excess fat & skin.



B. Liposuction of excess fat in the abdominal wall



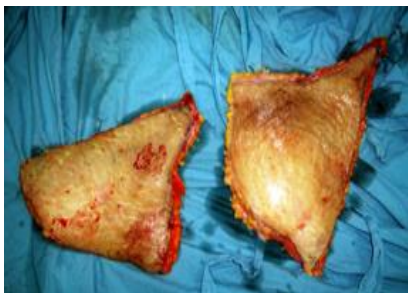
C. Elevation of the flaps – sparing the umbilicus



D. Tightening of the rectus muscles



E. Excess of the dissected flaps



F. Resection of the excess flaps



G. Approximation of the wound edges



H. Closure of the wound + steristrips + suction drain

MAMMAPLASTY

AUGMENTATION MAMMAPLASTY (BREAST ENLARGEMENT)

- **Indications:** Small, underdeveloped or asymmetrical breasts need to undergo breast augmentation using implants filled with saline/ silicon (Figure 13).
- **Objective:** This is a safe, effective surgical procedure designed to improve the contour of a woman's body by implanting FDA-approved implants under the pectoralis major muscle (sub-musculofascial) through an infra-mammary incision (Figure 14).
- **Anesthesia:** Local with sedation, or general.
- **Operative time:** 1-2 hours.
- It is usually done as an outpatient procedure.

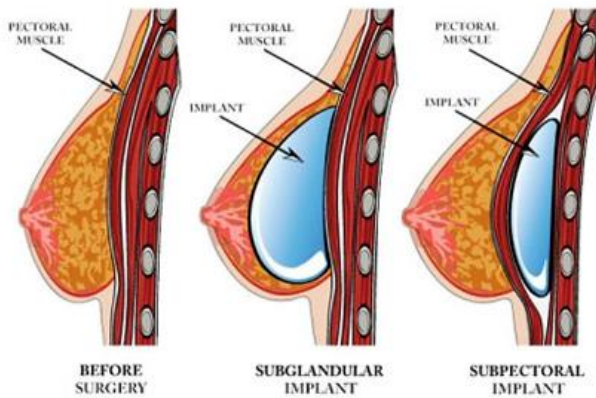


Figure 13. Breast augmentation implants (before & after implantation). The implant is usually placed sub-pectoral, but may be placed also sub-glandular.



Figure 14.
A. Intra-mammary incision,
B. Insertion of the implant to be spread inside (subperctoral or subglandular) after careful dissection,
C. Pre- and post-operative views.



- **Complications**
 1. Capsular contracture
 2. Infection
 3. Hematoma formation - bruising
 4. Changes in nipple sensation

MASTOPEXY (BREAST-LIFT)

- **Definition:** Mastopexy is an operation designed to remove the excess breast skin and reposition and resize the nipple-areola to produce an improved firmer and more uplifted breast appearance.
- **Indications:** *Breast ptosis* (hanging breast, sagging or drooping) due to:
 1. Weight loss.
 2. Pregnancy and lactation
 3. Aging
- **Procedure**
 - The parenchyma is positioned and supported upward by removing the skin from the area of the infra-mammary crease as well as vertically from beneath the nipple-areola (Figure 15).
 - This procedure produces an inverted-T scar (similar to that produced after reduction mammoplasty) as shown in Figure 16.
- **Anesthesia:** General anesthesia, or local with IV sedation.
- **Operative time:** 1-3 hours – performed as an in-patient procedure.

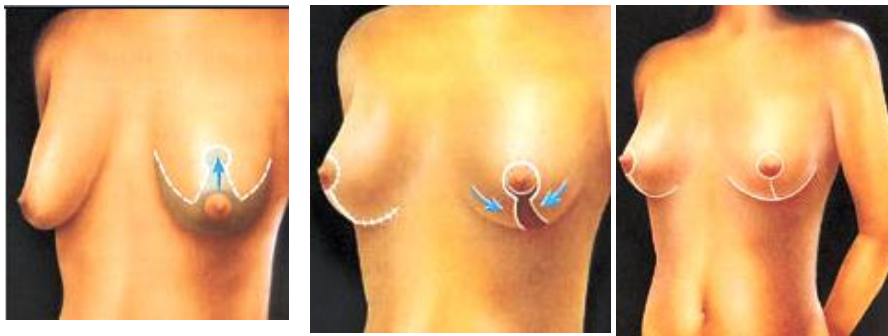


Figure 15. Steps of mastopexy (breast-lift)

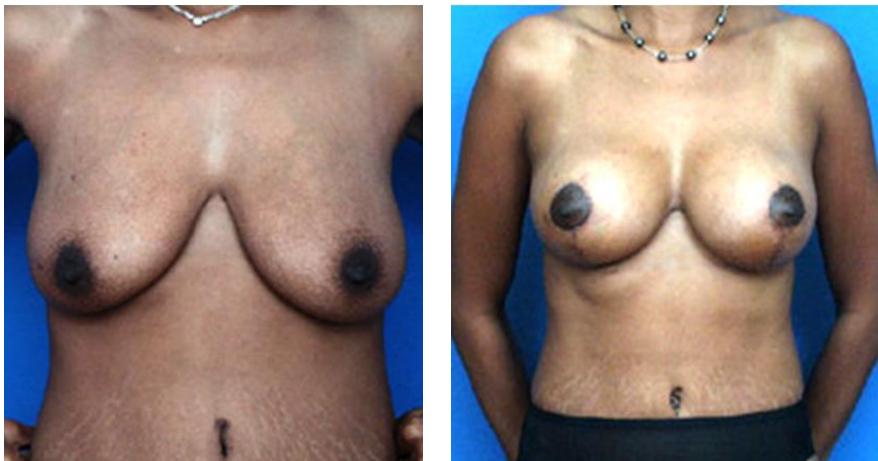


Figure 16. A lady with ptosed (sagging) breast (left) after mastopexy (breast-lift) (right). Note the inverted T-shaped scar.

- **Complications**
 1. Temporary bruising -swelling - hematoma
 2. Discomfort & numbness
 3. Dry breast skin.
 4. Permanent scars.

REDUCTION MAMMAPLASTY (BREAST REDUCTION)

- *Indications*
 - *Disfigurement*: Women with unusually large, sagging or uneven breasts are dissatisfied with their physical appearance.
 - *Pain or discomfort* from the sheer weight of their breasts & the pressure of brassiere straps on their shoulders. Large breasts can also hamper a woman's physical activities & make it difficult to find properly fitting clothes, particularly brassieres.
- *Types of reduction mammoplasty*
 - Amputation with free NA graft
 - Pedicled flap: Superior - inferior or central - bipedicled (horizontal or vertical) - medial -lateral.
- *Types of scars*: Inverted T (wise pattern) - vertical - lateral resection - circumareolar skin resection - horizontal with no vertical scar.
- *Procedure (inferior pedicle technique or brassiere pattern skin reduction)*
 - This technique involves both vertical & horizontal incisions made around the nipple area after which excess fat, tissue & skin are removed from the sides of the breast. These excisions, when brought together, result in an incision that resembles an inverted-T.
 - Following excision, the nipple, areola & tissue below are relocated but not detached.
 - Small sutures are used to close the incisions.

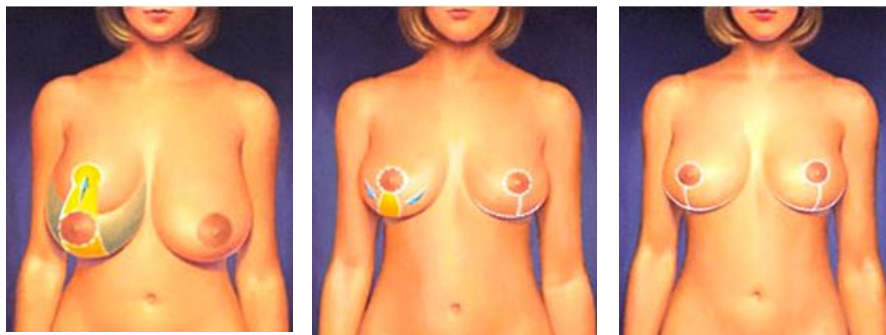


Figure 17. Steps of inferior pedicle reduction mammoplasty



Figure 18. A lady with large (sagging) breast (left) after reduction mammoplasty (right). Note the inverted T-shaped scar.

- *Complications*
 1. Nipple–areolar necrosis (most important)
 2. Hematoma or seroma formation
 3. Infection
 4. Wound healing defects.
 5. Under-resection.
 6. Asymmetry.

GYNECOMASTIA

Introduction

- *Definition:* Generalized enlargement of the male breast. It may be physiologic or pathologic
- Is common in men & can result in embarrassment, cruel teasing and social trauma.
- Men so affected will try to hide it with thick shirts, avoiding bare chested activities and withdrawing from public exposure.

Grades of Gynecomastia

Simon divided gynecomastia into 4 grades (Figure 19)

- Grade 1 : Small enlargement, no skin excess
- Grade 2a: Moderate enlargement, no skin excess
- Grade 2b: Moderate enlargement with extra skin
- Grade 3 : Marked enlargement with extra skin

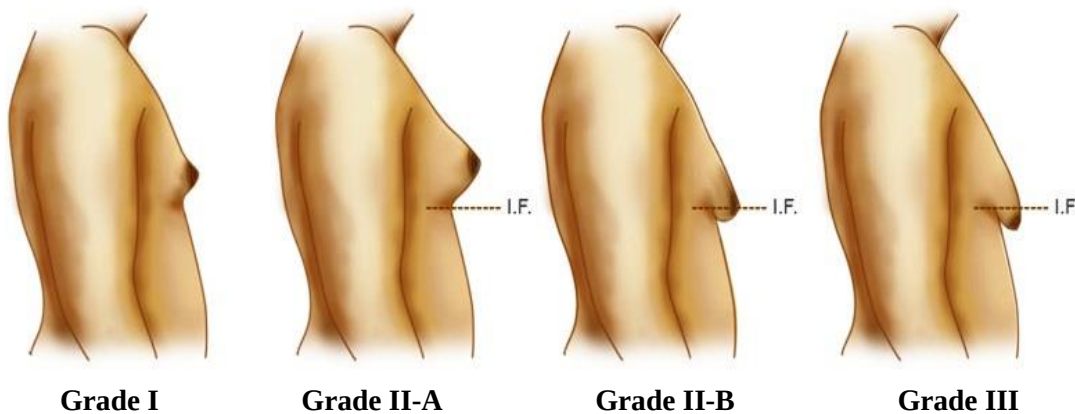


Figure 19.
Grades of
gynecomas-
tia (Simon)

Common Causes of Gynecomastia

1. Idiopathic - Obesity
2. Physiologic: Neonatal – pubertal (adolescent) – old age (senescence gynecomastia).
3. Endocrine
 - *Testis:* hypogonadism, Klinefelter syndrome
 - *Adrenal:* Cushing syndrome, congenital adrenal hyperplasia
 - *Thyroid:* hypothyroid, hyperthyroid
 - *Pituitary:* pituitary failure
4. Neoplasms: Adrenal – testicular – pituitary tumors – bronchogenic.
5. Systemic diseases: Renal failure – liver cirrhosis – adrenal disease – malnutrition.
6. Drug-induced
 - *Hormones:* Estrogens, androgens
 - *Anti-androgens:* Spironolactone, cimetidine, ketoconazole, ranitidine, flutamide
 - *Cardiovascular drugs:* Amiodarone, digoxin, nifedipine, reserpine, verapamil
 - *Abused drugs:* Alcohol, heroin, marijuana

Treatment

Physiological Gynecomastia

- It usually does *not* require treatment.
- Some pubertal cases may find it embarrassing and require *subcutaneous mastectomy* (preserving nipple and areola) or *Danazole* (anti-gonadotrophin, 200 mg b.d) (effective).

Pathological Gynecomastia

- Treatment of the underlying condition:
 - In *androgen deficiency*, testosterone administration may cause regression.
 - When it is caused by *medications*, then these are discontinued if possible.
 - When *endocrine defects* are responsible, then those patients receive specific therapy.
- Reversing gynecomastia with *Danazol* are successful, but with androgenic side effects.
- *Surgery* (SC mastectomy) is considered when gynecomastia is *progressive* and *does not respond* to other treatments. It is performed by a curved infra-areolar incision and the breast disk is excised after severing the ducts at the base of the nipple (Figure 20). Drainage is usually required. Nipple sensation is often lost but other complications are uncommon and include over-resection (results in a saucer-type deformity), under-resection, hematoma and infection.

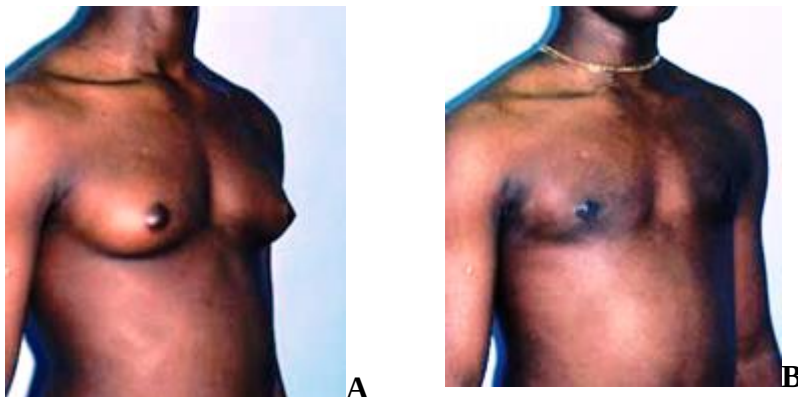


Figure 20. Surgical treatment of gynecomastia. **A:** pre-operative, **B:** post-operative

SKIN RESURFACING (NON-INVASIVE PROCEDURES)

1. LASER SKIN RESURFACING

- Laser stands for "light amplification by the stimulated emission of radiation."
- A CO₂ laser is used to remove areas of damaged or wrinkled skin, layer by layer. The laser beam can vaporize & remove wrinkles, scars & blemishes & can seal blood vessels.
- It is most commonly used to minimize the appearance of fine lines & pigmentation.
- What areas can be treated through laser resurfacing? The laser is used to reduce tiny wrinkles, acne scars & other minor skin imperfections, especially around the mouth & eyes, problems which often cause concern to both men & women (Figure 21).



Figure 21. Laser skin resurfacing of the lips (A: pre- and B: post-laser) of an old lady.

2. MICRO-DERMABRASION

- A Sandblaster-like device is used to spray high pressure stream of Aluminum oxide or salt crystals across face and suction is used to remove dead *outer layer* of skin.
- It stimulates skin cell and collagen production.
- It is used to reduce fine lines, “crow’s feet”, age spots and acne scars (Figure 22)
- It is an out-patient procedure (*lunch hour procedure*) that is effective for all skin types
- Multiple treatments are required for visible results; 5-12 treatments, 2-3 weeks apart
- Complications are few because the treatment extends only to outermost layer of skin and so scarring is unlikely and recovery is rapid.
- Use of eye protection during procedure prevents ocular complications such as redness, sensitivity to light and crystals adhering to the cornea typical.



Figure 22. Micro-dermabrasion of the face of a young gentleman with acne scars (pre- and post-procedure views).

3. CHEMICAL PEEL

- According to depth, chemical peel could be: superficial, medium or deep.
- It utilizes a solution of *Phenol* (deep peel), *Trichloroacetic acid* (medium depth peel) and *Alpha hydroxy acids* (light peel) to remove damaged outer layers of skin
- It is used to treat wrinkles (Figure 23), blemishes, uneven pigmentation or sun damage

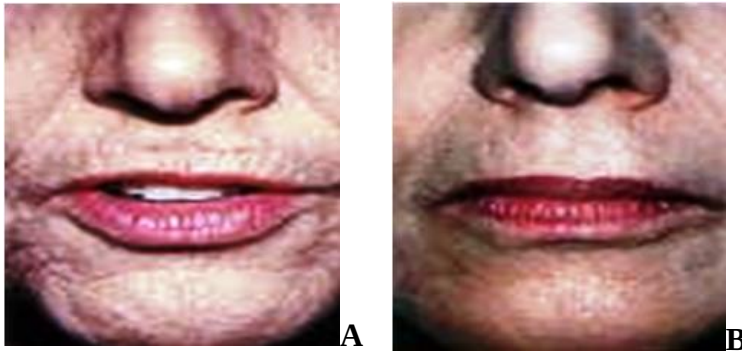


Figure 23. Chemical peel for treatment of wrinkles in an old lady (**A:** pre- and **B:** post-chemical peel).

- *Side Effects*
 1. Temporary throbbing, tingling, swelling, redness, sensitivity to sun, whiteheads
 2. With Phenol peel: permanent lightening of skin and loss of ability to tan.
- *Risks:* Infection, scarring, flare-up of skin allergies, fever blisters, cold sores.